
Part II

Orthopedics of the Upper Extremity

Hand and Fingers

6

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"Half crimping" in Halfway Log Dump, Ontario, Canada,
Photo Michael Simon

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6.1 Clinical Examination of the Hand

After taking a thorough history of the presenting injury, a detailed clinical examination is crucial to arriving at an accurate diagnosis. The history should focus on what kind of climbing move and finger position lead to the trauma and clarify whether the pain arose before, during or after climbing (e.g., tenosynovitis of the finger is usually painful before and after climbing; this injury may be pain-free during climbing if warmed up properly. A pulley strain is usually painful only under stress when climbing using the crimp position of the fingers).

The clinical examination both actively and passively tests the range of motion (ROM) of all three joints of the respective finger: the metacarpophalangeal (MCP), the proximal interphalangeal (PIP), and the distal interphalangeal (DIP). For the thumb, the ROM is tested for the basal joint, MCP joint, and IP joint. The function of the flexor tendons, flexor digitorum superficialis (FDS), and flexor digitorum profundus (FDP) needs to be tested separately (Figs. 6.1 and 6.2). The collateral ligaments of the MCP, PIP, and DIP joints need to be tested both in extension and 30° flexion in order to assess the proper and accessory components of each ligament (Fig. 6.3). The palmar plate of the PIP joint is assessed through the so-called “translation test,” which is similar to the Lachman test in the knee joint (Fig. 6.4). For the detection of flexor tendon



Fig. 6.1 Examination of the superficial flexor tendon (m. flexor digitorum superficialis) (FDS) (Photo M. Simon)



Fig. 6.2 Examination of the deep flexor tendon (m. flexor digitorum profundus) (FDP) (Photo M. Simon)



Fig. 6.3 Examination of the collateral ligaments of the PIP in 30° flexion (Photo M. Simon)



Fig. 6.4 Translation test of the palmar plate (Photo M. Simon)

bowstring phenomenon in a flexor pulley injury, the respective finger undergoes flexion against resistance. The patient is asked to flex the injured finger against their own thumb, while the clinician palpates and observes for clinical flexor



Fig. 6.5 Examination of a bowstringing phenomenon (Photo M. Simon)

tendon bowstringing in the palmar direction (Fig. 6.5). A palmar-sided tenderness to palpation at the finger's proximal phalanx is suggestive of an A2 pulley injury or inflammatory tenosynovitis. A laterally located palpation tenderness at the same phalanx is suggestive of a lumbrical muscle injury. The “lumbrical shift test” is performed for a specific examination of the lumbrical tendons (Fig. 6.6). A dorsally located point tenderness at the PIP joint is frequently found in epiphyseal fractures in adolescent climbers or joint capsulitis (articular synovitis) in skeletally mature climbers. To detect hamulus fractures a specific stress test is performed (Fig. 6.7). Finger flexion in ulnar abduction causes shear forces onto the hamulus, provoking pain.

6.2 Diagnostic Imaging

Many acute and chronic finger conditions require plain radiographs to confirm the diagnosis and/or rule out other possible diagnoses. However, in recent years, dynamic musculoskeletal ultrasound has become the “workhorse” imaging technique for climbers’ finger injuries. Ultrasound assessment is very user-dependent and therefore may not be easily available to all healthcare providers. In many European countries, ultrasound examinations are performed routinely by orthopedic surgeons or sports medical doctors themselves. However, in other countries such as the UK, Canada, and certain regions in the USA, these exams are performed by either



Fig. 6.6 Lumbrical shift test: an extended middle finger is stressed while the neighboring fingers are bent, thus leading to a stress of the lumbrical muscles common body (Photo M. Simon)



Fig. 6.7 Hamulus stress test: finger flexion in ulnar abduction causes shear forces onto the hamulus, provoking pain (Photo M. Simon)

ultrasound technicians or radiologists. Consequently, some of the recommendations given within this book, especially in the chapter on hand injuries, may be difficult to implement in other countries. In the diagnosis of pulley injuries, tenosynovitis, or ganglion cysts, ultrasounds are very cost-effective and accurate with high sensitivity and specificity [1–4]. The technique of finger ultrasound examination is easy to learn and very efficient [5]. We predominantly use small, linear transducers (“hockey stick”) with 10–18 MHz in longitudinal and transversal planes (Fig. 6.8). For signal enhancement, either a gel standoff pad or normal ultrasound gel is used. The ultrasound is performed statically and dynamically; for proper detection of inflammation, a color-coded duplex ultrasound evaluation is used. In diagnosing pulley injuries, the dis-



Fig. 6.8 Ultrasound examination with an 18 Mhz linear transducer (Hockey stick) (Photo M. Simon)



Fig. 6.9 Ultrasound in “forced flexion” (Photo M. Simon)

tance between the flexor tendons and the bone (TB = tendon-bone distance) is measured under “forced flexion” [6] (Fig. 6.9). The forced flexion is achieved by asking the patient to use a healthy finger to press against the fingertip of the injured finger, which undergoes active flexion during examination. This causes stress and tension on the flexor tendons of the injured finger, increasing a possible TB in an injury. The TB is measured at the midportion of the base phalanx for the A2, at the midportion of the middle phalanx of for the A4, and at the level of the palmar plate for the A3 pulley. In the last case, the palmar plate-tendon distance is measured, as the A3 pulley inserts into the palmar plate, not the bone [7, 8]. An additional MRI or CT scan is only required in rare cases [9–18].



Fig. 6.10 Cortical hypertrophy of a 28-year-old climber as a stress adaptation (normal finding)

6.2.1 Normal Findings

The high impact of rock climbing onto the fingers leads to physiological adaptations over the years. These adaptations must be strictly separated from pathological changes [19]. Through radiographic and MRI analyses, Hochholzer et al. [15] and Heuck et al. [20] identified these adaptations. The authors found an adaptive hypertrophy of the joint capsules, thickening of the collateral ligaments, cortical hypertrophy, and a hypertrophy of up to 50% of the flexor tendons themselves (Fig. 6.10). Hochholzer and Heuck also showed that certain reactions, such as cortical hypertrophy, are likely protective adaptations to the high stress of the sport rather than early pathological osteoarthritic

changes. Therefore, clinical and radiographical findings must be interpreted carefully. In an analysis of radiographs in young high-level climbers as well as 140 experienced climbers, Schöffl et al. [21] classified the following findings as stress reactions, using a modified Kellgren Lawrence classification [22]: cortical hypertrophy, subchondral sclerosis/increased thickness of epiphysis, calcifications of the insertion of the flexor digitorum superficialis or the flexor digitorum profundus tendon, broadened proximal, and/or distal interphalangeal joint base.

6.2.2 Distribution of Finger Injuries

Table 6.1 summarizes the distribution of diagnoses in our own patients presenting with finger pain over three different time periods.

In the following sections, we discuss various specific finger injuries. In general, distinction should be made between acute traumatic finger injuries (e.g., pulley injury, tendon strain, ligamental injury, soft tissue injury, etc.) and chronic overstrain injuries (e.g., tenosynovitis, osteoarthritis, epiphyseal growth plate injury, lumbrical shift syndrome, etc.).

6.3 Finger Injuries

6.3.1 Pulley Injuries

An injury to the finger flexor tendon pulleys is the most common climbing-specific injury. Closed traumatic ruptures of finger flexor tendon pulleys arose as a new complex trauma in the middle of the 1980s. Bollen [23–25] and Tropet et al. [26] were the first to observe this type of injury; subsequently, much research has

been carried out. In the past, the clinical management recommendations varied widely for these injuries. Currently, there is more consensus with little variation in treatment regimens between various authors [27]. Since this injury has become more well-known in medical literature, it has also been described in other sports [28] as well as in nonathletes and children [29]. Pulley injuries occur not only in the well-known annular (A) pulleys but also as isolated injuries of the cruciate (C) pulleys [30].

6.3.1.1 Clinical Presentation

Most climbers report an acute onset of pain while performing a hard move with the finger in crimp position. There may be a slip of the foot from a foothold, resulting in sudden loading of the finger. In some cases, a loud popping noise is heard. Occasionally, the rupture can occur due to chronic repetitive trauma and fatigue failure. In cases of prior chronic tenosynovitis treated with steroid injections, the pulley can rupture even in cases of minor stress if the appropriate rest after the last injection was neglected. Biomechanical analyses showed that pulley ruptures are mostly caused by eccentric finger movement [31].

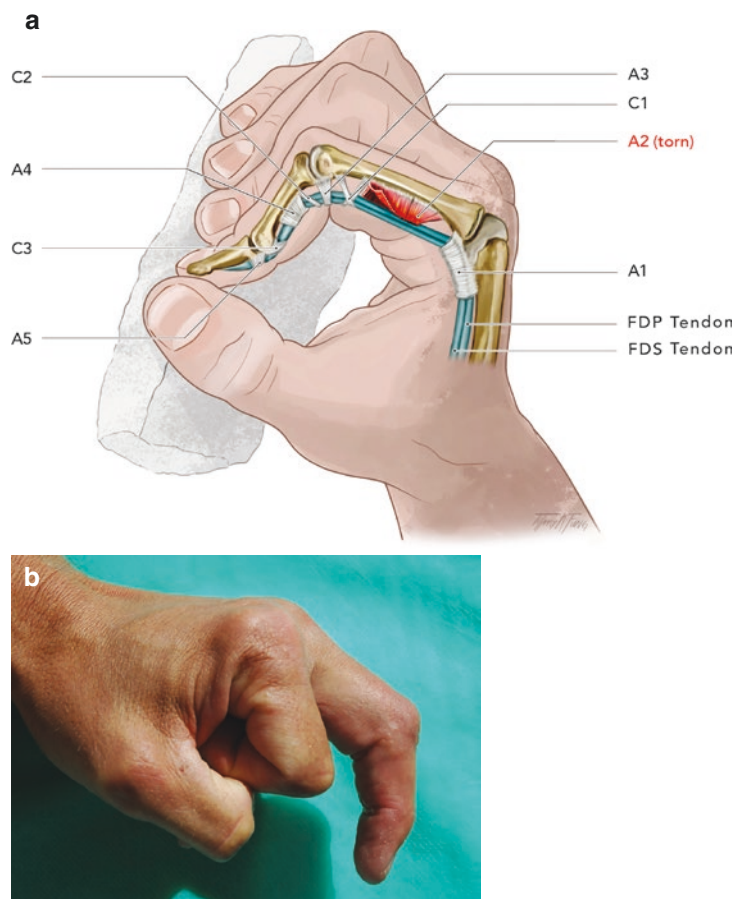
Physical examination reveals tenderness over the palmar level of the injured pulley accompanied by swelling and occasionally hematoma. A visible bowstring only occurs with multiple pulley ruptures (Fig. 6.11). In cases of A4 or A2 pulley rupture, increased distance of the flexor tendon to the bone can be palpated occasionally if the injured finger is opposed to the thumb in a crimping position (i.e., palpable bowstring) [23]. Since a definitive diagnosis cannot be made based on clinical examination exclusively, we always perform a dynamic ultrasound assessment. In cases of special questions or diagnostic uncertainty, an MRI can be helpful [7, 17, 32].

Table 6.1 Most frequent finger injuries 2017–2018 (*n* = 261), 2009–2012 (*n* = 474), and 1998–2001 (*n* = 247) [109]

Finger injuries 2017–2018 (<i>n</i> = 261) (2 years)	<i>n</i>	%	Finger injuries 2009–2012 (<i>n</i> = 474) (4 years)	<i>n</i>	%	Finger injuries 1998–2001 (<i>n</i> = 247) (4 years)	<i>n</i>	%
Pulley injury	78	29.9	Pulley injury	140	29.5	Pulley injury	122	49.4
Tenosynovitis flexor tendon	67	25.7	Capsulitis	87	18.4	Tenosynovitis	42	17.0
Capsulitis	49	18.8	Tenosynovitis flexor tendon	80	16.9	Strain finger joint capsule	37	15.0
Epiphyseal fracture	19	7.3	Strain flexor tendon	36	7.6	Capsulitis	13	5.3
Lumbrical tear/strain	12	4.6	Strain finger joint capsule	25	5.3	Ganglion	11	4.5
Strain finger joint capsule	10	3.8	Ganglion finger flexor tendon	19	4.0	Strain flexor tendon	7	2.8
Osteoarthritis	5	1.9	Lumbrical shift syndrome	19	4.0	Fracture	7	2.8
Strain flexor tendon	4	1.5	Collateral ligament injury	17	3.6	Osteoarthritis	7	2.8
Ganglion finger flexor tendon	3	1.2	Epiphyseal fracture	16	3.4	Soft tissue injury	5	2.0
Contusion	3	1.2	Osteoarthritis	14	3.0	Tendon rupture	4	1.6
Phlegmonia/cellulitis	2	0.7	Extensor hood syndrome	7	1.5	Collateral ligament injury	3	1.2
Collateral ligament injury	2	0.7	Lumbrical tear/strain	4	0.8	Osseous tear fibrocartilago palmaris	2	0.8
Distorsion thumb	1	0.3	Snap finger	3	0.6	Epiphyseal fracture	2	0.8
Disruption volar plate	1	0.3	Cartilage injury	2	0.4	Lumbrical tear/strain	2	0.8
Cartilage injury	1	0.3	Flip phenomena	2	0.4	Phlegmonia/cellulitis	1	0.4
Neuropaxia	1	0.3	Broken osteophyte	1	0.2	Finger amputation	1	0.4
PIP joint dislocation	1	0.3	Avulsion fracture	1	0.2	–	–	–
Snap finger	1	0.3	Flexor contraction	1	0.2	–	–	–
Contracture finger flexor tendon	1	0.3	Rupture connexus intertend.	1	0.2	–	–	–
–	–	–	Enchondroma	1	0.2	–	–	–
–	–	–	Contusion	1	0.2	–	–	–
–	–	–	Tendon rupture	1	0.2	–	–	–

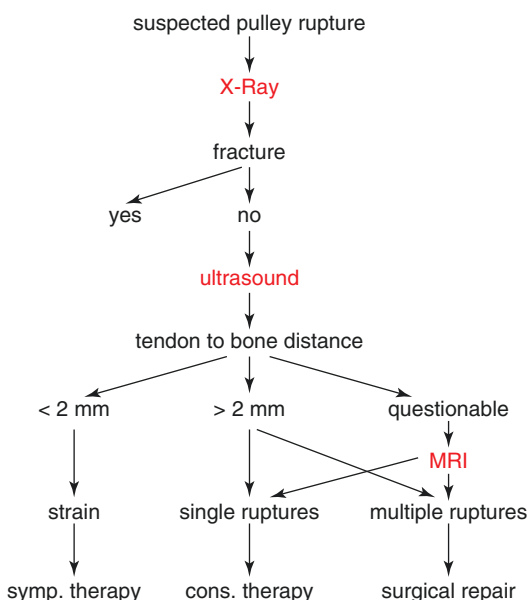
Fig. 6.11 Bowstringing in a A2 pulley rupture.

(a) biomechanical drawing, (b) clinical presentation (Pic by Tiffany Fung)



6.3.1.2 Diagnostic

The diagnostic follows our algorithm [33]:



According to the algorithm as well as the reports of Gabl et al. [34], radiographs should be performed to exclude fractures or osseous tears of the palmar plate as well as chronic repetitive stress epiphyseal fractures in adolescent climbers, if warranted by the clinical picture. For further evaluation, the “gold standard” is the dynamic ultrasound. MRI is only performed if the X-rays and ultrasound do not provide a conclusive diagnosis (Fig. 6.12).

The advantage of the ultrasound is the fact that it can be performed dynamically, which is only possible with an MRI in very specific studies [7, 16]. A direct visualization of the various pulleys is often possible with a high-resolution ultrasound probe (15–18 MHz, “hockey stick” probe) (Fig. 6.13). Valid detection of the tendon-bone distance (TB) under “forced flexion” can also always be performed with a lower resolution probe, even with some of the new handheld ultrasound devices [1] (Fig. 6.14). Regarding the TB



Fig. 6.12 MRI in a A2,3,4 pulley rupture

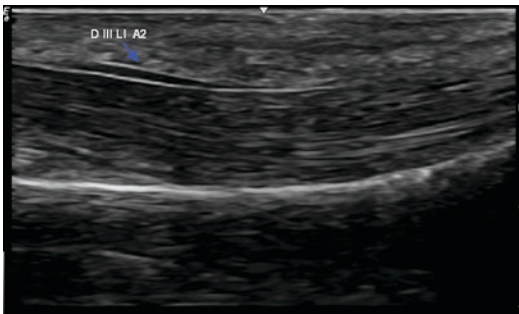


Fig. 6.13 Normal A2 pulley in the ultrasound (the flexor tendon follows the course of the bone; a direct visualization of the pulley is possible)

distance in a noninjured population, Bassemir et al. [35] described the normal distances in an analysis of 200 non-climbing individuals.

The following cutoffs could be evaluated in our cadaver study for pulley injury detection under “forced flexion” of 10 N [7] and are our clinical guidelines:

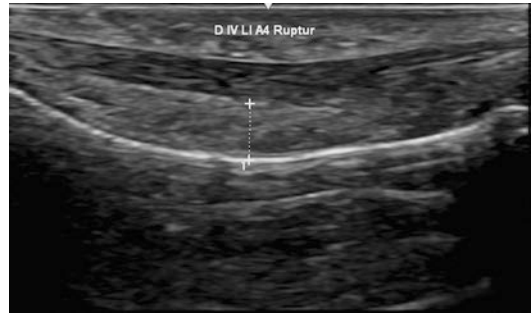


Fig. 6.14 A4 pulley rupture (the flexor tendons are distant to the bone)

Ultrasound criteria for pulley tear under forced flexion of 10 N [7]:

- **A2: TB > 2 mm: sensitivity 94%, specificity 100%**
- **A3: VP > 0.9 mm: sensitivity 76%, specificity 94%**
- **A4: TB > 2 mm: sensitivity 90%, specificity 97%**

(TB = tendon-bone distance, VP = volar plate distance)

For the A2 and the A4 pulley, the TB distance is measured at the midportion of the respective phalange; for the A3, the distance tendon to volar plate is measured. To get the correct measuring point of the volar plate distance (VP) distance, the length of the VP is measured from its most proximal to its most distal point. A point is then defined on the proximal phalanx where the curvature of the condyle transitions from concave to convex. Then, the distance is measured between this point and the middle of the volar aspect of the VP (VP bone). Next, the distance is determined between this point (measuring base point) and the tendon in extension to obtain the VP tendon distance [7] (Fig. 6.15).

Clear TB distance cutoffs for single C pulley injuries do not exist, as the TB distance is only slightly increased. For the diagnosis of C pulley injuries, a combination of the detection of hematoma, a change of the tendons course in the dynamic ultrasound examination, and a minor increase in TB distance lead to the final diagnosis.

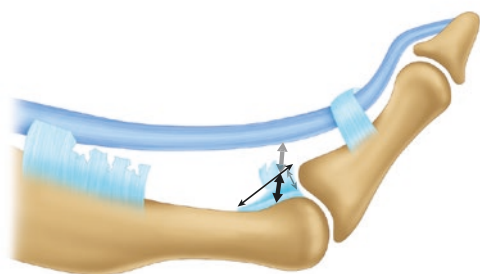


Fig. 6.15 Stylized drawing of a finger after pulley injury of the A2 and A3 pulleys with the measurements taken during the study (thick black arrow = VP bone distance, thick grey arrow = VP tendon distance, thin black arrow = VP length, thin grey arrow = VP thickness) [7]

Multiple complete pulley ruptures, especially triple-pulley ruptures (i.e., A2/3/4) can cause clinically visible bowstringing and a functional deficit, as the longitudinal force of the flexor tendons can no longer be transferred into a rotation of the finger joints (DIP, PIP). Without treatment, this will often lead to chronic flexion contractures, usually at the PIP joint (Fig. 6.16).

6.3.1.3 Treatment

Single pulley injuries (grade I–III) receive conservative therapy with an optional initial short period of immobilization (depending on the grade of swelling and collateral damage) and early functional therapy with a thermoplastic pulley support ring (Figs. 6.17 and 6.18) [36]. The design for these pulley support rings presented by Schneeberger et al. [37] is the best, as these are the only ones, which allow for anatomic repositioning of the flexor tendons without compression of the neurovascular bundle. Circular rings cannot relocate the tendon close enough to the bone without compromising the digital neurovascular structures. With anatomic repositioning of the tendons, the injured pulley, which generally does not tear in the middle but



Fig. 6.16 Contracture as a result of an untreated A2,3,4 pulley rupture

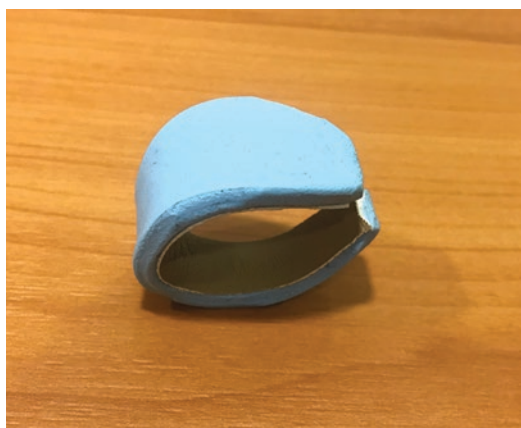


Fig. 6.17 Pulley support ring (fixed with a circular tape)

from one side of its insertion [38, 39], can heal while forming a stable scar (neo-pulley). An ultrasound evaluation is essential to exclude a FLIP-phenomenon (flap irritation phenomenon), in which the torn pulley slips underneath the flexor tendons. This prevents relocation of the tendons and produces chronic irritation of the tendons, resulting in tenosynovitis. Occasionally, a FLIP-phenomenon requires a secondary surgical pulley repair [40, 41]. The initial immobilization is only rarely necessary; the pulley support ring should be worn for 6–8 weeks in a single pulley tear [37]. For a grade 1 injury, pulley strain taping is usually sufficient. When wearing the pulley support ring, it is crucial to maintain proper skincare and daily application of nourishing ointments

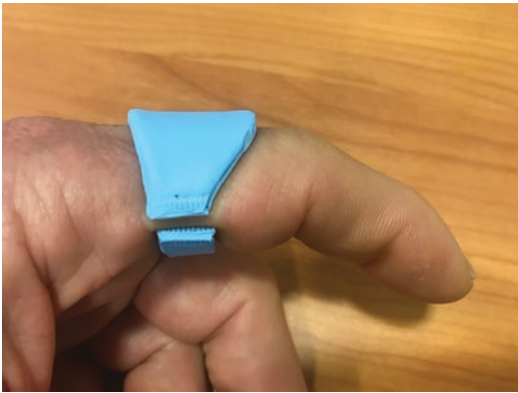


Fig. 6.18 Pulley support ring (fixed with a circular tape)

(e.g., panthenol) to prevent pressure sores (Fig. 6.19). A coating of the inner layer of the pulley support rings with a thin layer of leather can reduce skin ulcerations. The results of this conservative therapy are mostly excellent; the former increased TB distance is permanently reduced and the initial strength in finger flexion will resolve as the climbers regain their pre-injury strength level [42]. Following the initial pulley protection ring, H-taping of the finger should be applied during the “return-to-sport” period for up to 6–12 months [43].

In grade IV injuries (multiple pulley injuries), the general recommendation is a surgical repair [36]. Otherwise, the permanently increased TB distance and scarring will cause flexion contracture [27]. If this contracture is already present, a later, secondary surgical reduction is challenging. In recent years, we have modified our original therapeutic and grading system [36, 44], as recent studies also showed a good outcome of conservative management in certain types of multiple pulley ruptures without clinical bowstringing [27, 37]. Thus, we divided grade IV injuries into the subgroups of grade IVa and grade IVb injuries. In some cases, grade IVa injuries can be treated with conservative therapy [36]. Grade IVa injuries represent A2/A3 or A3/A4 injuries with no major clinical bowstringing, an ultrasound-proven possibility of repositioning the flexor tendons to the bone, a therapy start of less than 10 days after the injury and no contracture. A2/A3 or A3/A4 injuries with obvious bowstringing are categorized as



Fig. 6.19 Pressure skin ulceration caused by a pulley support ring

grade IVb injuries, as are all other multiple pulley injuries (see Table 6.2) [36].

Nevertheless, it is important to mention that the decision toward a conservative management in a multiple tear should always be made specifically for each individual case. This decision should be made in consultation with the patient and based on careful analysis of the clinical findings, including a proper ultrasound diagnosis. Additionally, close patient follow-up is necessary (including ultrasound controls), and surgical repair must immediately be performed if contracture begins to develop. In many ways, conservative therapy for a multiple pulley injury is actually more challenging, than surgical repair, and should only be undertaken by experienced clinical teams at high-volume centers. Conservative therapy of double a pulley rupture is only possible if the therapy starts promptly after the injury (ideally within one to 2 weeks), if flexor tendons can be anatomically repositioned under ultrasound monitoring with no FLIP-phenomenon [41], and if no flexion contracture has developed

Table 6.2 Therapeutic guidelines of pulley injuries in rock climbers (Consensus Statement of Expert Commission Sportsmedicine Bamberg (Volker Schöffl, Isabelle Schöffl, Christoph Lutter, Michael Simon, Yasser El-Sheikh), Bamberg 2019 [36], modified after (Schöffl et al. 2004c))

	Grade I	Grade II	Grade III	Grade IV a	Grad IV b
	Pulley strain	Complete tear of A3 or A4, partial tear of A2	Complete tear of A2	Multiple ruptures: – A2/A3 or A3/A4 rupture if: – No major clinical bowstring – Ultrasound-prooven possibility of reposition of the flexor tendon to the bone – Therapy starting <10 d after injury – No contracture	Multiple ruptures: – A2/A3 or A3/4 with obvious clinical bowstring – A2/A3/A4 rupture – Singular pulley rupture with FLIP phenomenon – Singular rupture with increasing contracture – Singular rupture with secondary, therapy-resistant, tenosynovitis
Therapy	Conservative	Conservative	Conservative	Conservative, if secondary onset of PIP contracture >20° secondary surgical	Surgical
Immobilization	None	optional, <5 days	Optional, <5 days	Optional, <5 days	Postsurgical 14 days
Functional therapy with pulley protection (defined)	2–4 weeks H-tape (during day time) or thermoplastic ring	6 weeks thermoplastic pulley ring	6–8 weeks thermoplastic pulley ring	8 weeks thermoplastic pulley ring	4 weeks thermoplastic ring (after 2 weeks of immobilization)
Easy sport-specific activities	After 4 weeks	After 6 weeks	After 8 weeks	After 10 weeks	After 4 months
Full sport-specific activities	After 6 weeks	After 8–10 weeks	After 3 months	After 4 months	After 6 months
H-taping during climbing	3 months	3 months	3 months	>12 months	>12 months

and a proper pulley ring is applied, keeping the tendons close to the bone [27]. The pulley support ring is worn for 23 h per day and is only removed once daily for skin care.

If a significant secondary flexion contracture of more than 20° extension deficit occurs, a surgical revision and pulley reconstruction is recommended. A minor contracture of up to 5° often develops and can be accommodated. In complex rupture situations (multiple pulleys (grade IVb) or after failure of a conservative therapy in grade IVa injuries, surgical repair in the form of a pulley reconstruction is necessary. In these cases, the

clinical presentation is majorly compromised movement and reduced flexion strength of the DIP joint. If untreated, secondary contracture at the PIP will follow [41].

Suture repair of the torn pulley at any stage is rarely possible, due to primary and secondary contracture of the ligament, with shortening and friability of the tissues preventing reliable end-to-end approximation, so pulley reconstruction is necessary. Out of the many various pulley reconstruction techniques (including the “Kleinert and Bennet” repair based on Weilby’s idea [45], Karev’s “belt-loop” technique [46], “single-loop”

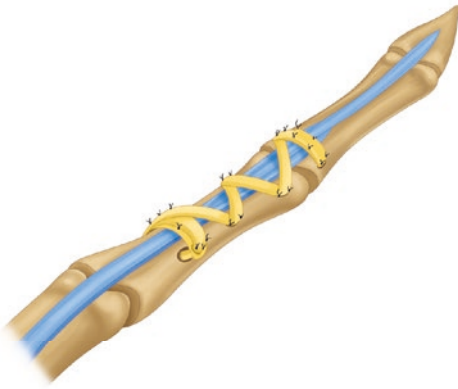


Fig. 6.20 Transosseous pulley repair [57]

technique according to Bunnel [47], “Lister repair” with retinaculum flexorum [48], palmaris longus tendon transplantation through the volar plate according to Doyle [49], Widstrom’s “loop-and-a-half” technique [50], “triple-loop technique” by Okutsu [51], or “extensor retinaculum graft” by Gabl [52] and Moutet [53]), we favor a modified “loop-and-a-half” technique with a distal continuation to the A3 pulley [54]. This combines the advantages of the “loop-and-a-half” as the strongest and the Weilby repair as the most functional repair [50, 54, 55]. We recently noticed that some osseous necrosis of the phalanx can occur following this repair, likely due to the high pressure of the circulation onto the bone and its blood vessels [56]. Therefore, we have modified our procedure into a transosseous repair (Figs. 6.20 and 6.21) [57]. A FLIP-phenomenon, an interposition of the pulleys stump underneath the tendons and the bone, as shown in Fig. 6.22 (Fig. 6.22) can render a conservative therapy impossible and require surgery. This can either be performed using the palmaris longus or a retinaculum extensorum flap [40].

Postsurgically, the finger is immobilized until completion of skin healing, using a palmar-sided cast splint for a maximum of 2 weeks. Thereafter,

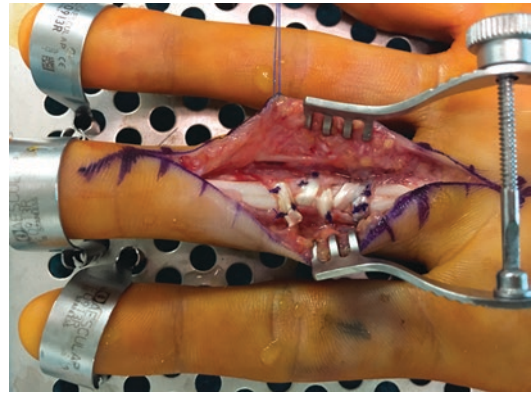


Fig. 6.21 Transosseous pulley repair with a free palmaris longus graft during surgery

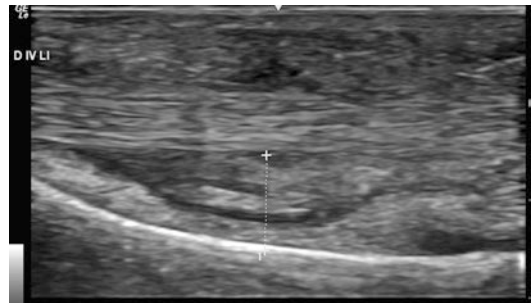


Fig. 6.22 A2 pulley rupture with an increased tendon-bone distance (see the line in the figure) and the trunk of the former pulley squeezed underneath the flexor tendons as a “FLIP”-phenomenon

a thermoplastic pulley support ring is implemented and gentle active ROM exercises commence. Six weeks after surgery, free movement without the support ring is allowed, and the functional therapy can be intensified. Specific rehabilitation exercises are presented in “One Move Too Many” (Schöffl V, Hochholzer T, Lightner S Jr) [58]. General exercise (e.g., light running) can be resumed after 2 weeks, once the surgical site edema has completely resolved. Easy climbing and climbing-specific training are allowed after 4 months (with protective H-taping) (Figs. 6.23, 6.24, 6.25, and 6.26). Taping should be continued through climbing for another 6 months. The full, sport-specific load can be assumed after 6–9 months.

Flexor tendon deflection in the crimping position

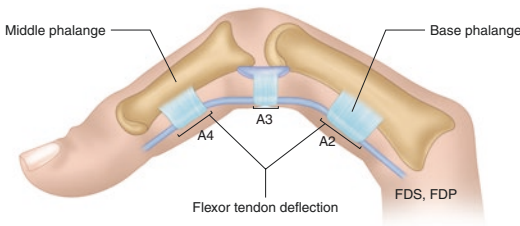


Fig. 6.23 Biomechanics of the H-tape. Based on the support of the A3 pulley, the stress onto the distal rim of the A2 pulley and the proximal rim of the A4 pulley is reduced, as the angulation of the flexor tendons is decreased. This helps in a tenosynovitis (see Sect. 5.3.2) and in a pulley injury [43]

Reduced flexor tendon deflection in the crimping position with H-tape

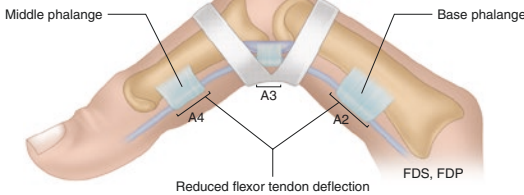


Fig. 6.24 Biomechanics of the H-tape. Based on the support of the A3 pulley, the stress onto the distal rim of the A2 pulley and the proximal rim of the A4 pulley is reduced, as the angulation of the flexor tendons is decreased. This helps in a tenosynovitis (see Sect. 5.3.2) and in a pulley injury [43]

6.3.1.4 Outcome

Schneeberger et al. [37] showed that through consistent use of the pulley support ring, the increased TB distance can be reduced. Also, the surgical management of multiple pulley ruptures showed good to excellent sport-specific results in our sport-specific pulley injury outcome score (see Chap. 2) [54] (Figs. 6.27 and 6.28). This is in agreement with the current literature, which shows good outcomes after pulley reconstruction in general, with no significant difference in outcomes between techniques [32, 52, 54, 59]. In an

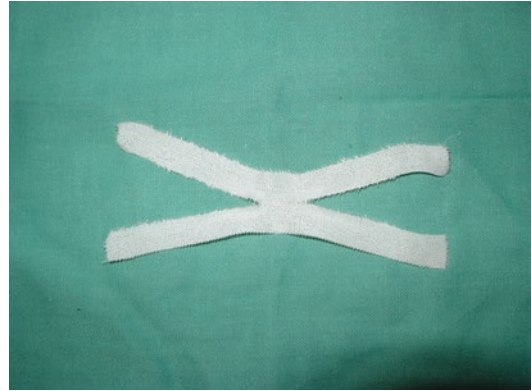


Fig. 6.25 H-taping



Fig. 6.26 H-taping



Fig. 6.27 Three-month postsurgical result of the patient in Fig. 6.21

outcome analysis, Schöffl et al. found a “return to climbing” in all of the 81 conservatively treated patients (grade I–III); only one patient needed a



Fig. 6.28 Three-month postsurgical result of the patient in Fig. 6.21

secondary pulley reconstruction due to a chronic tenosynovitis [44]. In another study concerning finger strength after conservative treatment of single pulley injuries, no patients showed a remaining strength deficit [42]. Nevertheless, in both groups (conservative and surgical), a minor extension deficit (5° – 7° contracture) at the PIP joint frequently remained. However, this usually has no clinical impact. If patients present for a pulley repair with a preexisting contracture due to an older pulley injury, we start with conservative dynamic extension splint therapy and aim to reduce the contracture to 20° prior to surgery. If this is not possible, we release the joint capsule surgically in addition to the pulley repair; however, minor permanent contracture of up to 20° must be expected.

6.3.1.5 Prevention

Many climbers try to protect their flexor tendon pulleys by applying circumferential tape around the proximal phalanx [27]. However, while some reduction of the bowstring is achieved via tape in cases of a disrupted pulley, no effect can be expected from protective pulley tape around an intact pulley [60, 61]. Taping over healthy pulleys did not show any biomechanical effect in a cadaver study [62]. Woollings et al. [63] even reported that preventive taping increases the risk for finger injuries. The main positive effect of prophylactic taping may be that the PIP joint is not flexed more than 80° – 90° if the tape is applied close or even over the PIP joint itself [27]. A cor-

rect warming-up procedure and the avoidance of a pronounced crimp grip position seem to be more important factors in pulley injury prevention [27]. It has been shown that over the first 100 to 120 climbing moves, the amount of physiological bowstringing of the flexor tendons shows an increase of up to 30% [64]. This effect could only be shown with performing climbing movements and was not observed with other warming-up techniques [64]. Therefore, it is recommended to warm-up by climbing about 3–4 routes with 40 moves or 8–12 boulder problems [64] with increasing intensity. After a pulley injury, tape should be applied at either the distal end of the respective injured pulley [64] or as an H-tape at the level of the PIP joint [43] for 6–9 months (see Chap. 22. Taping) (Fig. 6.29). Another potential preventive measure for finger pulley injuries is controlled low-intensity eccentric training of the finger flexor muscle-tendon units, which is shown to help in tendon healing [65] and may also have some effect on the pulleys [66]. Nevertheless, scientific evidence for this is still pending.

- **History:** Mostly one hard move in a crimping position
- **Onset:** Sudden
- **Clinical:** Pressure tenderness palmar-sided at the pulley, often a “popping” sound during the event
- **Diagnostic:** Clinical findings, ultrasound, optional MRI
- **Therapy:** Singular pulley injuries conservative, multiple mostly surgical
- **Outcome:** Good, no consistent strength deficit

6.3.2 Finger Flexor Tenosynovitis

Inflammatory flexor tenosynovitis (aka: tendonitis, tendovaginitis) is inflammation of the flexor tendon sheath and is the most frequent overuse syndrome in climbers’ fingers [67]. An inflammatory response occurs after repetitive stress, and its onset can be acute after one exceptionally hard training or climbing day, or creeping up

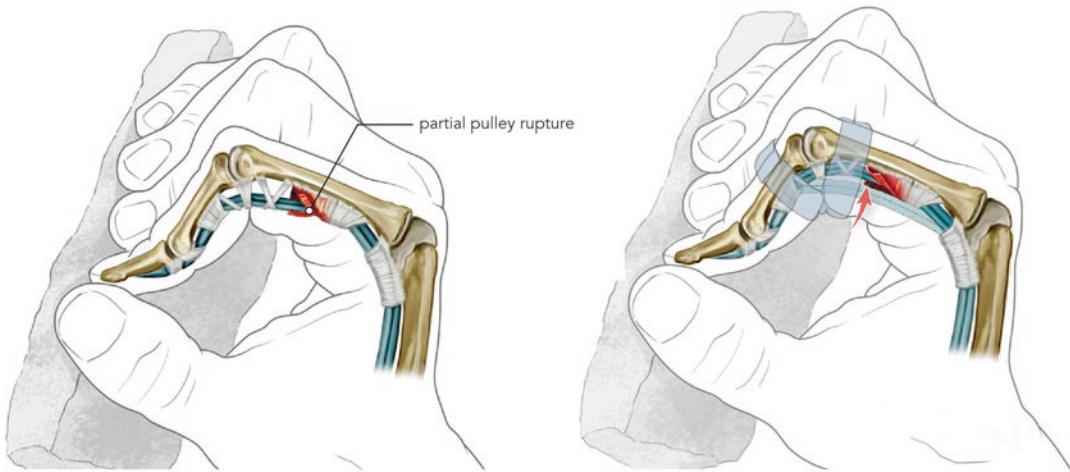


Fig. 6.29 H-taping in partial A2 tear. (Pic by Tiffany Fung)

slowly over days [19]. The latter condition is commonly referred to by laypeople and climbers as tendonitis but in fact represents an inflammation of the tendon sheath. Thus, the correct term is “tenosynovitis” [6, 19, 67].

The etiology of inflammatory tenosynovitis in climbers is usually repetitive high stress while performing a so-called “crimping” position [19]. In a crimping position, the flexor tendons are curved at the distal end of the A2 finger flexor tendon pulley, as well as the proximal end of the A4 pulley [67–69]. This causes an increased flexion of the tendons and thus increased friction onto the rim of the pulley [43, 69]. Repetition of this position can lead to a chronic inflammatory reaction of the tendon sheath [19, 67]. Even though no studies have been performed on a histopathological level, ultrasound examination of climbers with this condition, including duplex sonography, revealed an inflammatory reaction within the flexor sheath and its lining [1, 70, 71]. While there are generally multiple extrinsic and intrinsic factors in the origin of tenosynovitis, the main pathological stimulus for this condition is considered to be the excessive loading of tendons during vigorous training as performed in rock climbing [72]. Tendons respond to repetitive overload beyond the physiological threshold either with inflammation of the sheath, degeneration of the tendons, or both [73]. Tendon damage may even occur from stress within the physiological limits, as frequent

cumulative microtrauma may not allow enough time for the tendons to heal [73], a case that is highly probable in ambitious climbers [67]. In general, no other sport than climbing places such an immense load onto the fingers. The frequent use of the crimping position in climbing is increasing this load additionally [68, 69].

6.3.3 Clinical Presentation and Diagnostic

The climber suffers from pain, occasionally accompanied by a minor swelling along the palmar surface of the finger, just around the same area as in a pulley injury. The pain can extend into the palm or the forearm [67] (Fig. 6.30). The clinical examination usually presents point tenderness at the level of the inflammation, mostly at the A2 or A4 pulley. Sometimes, the swelling of the tendon sheath is palpable due to the increased fluid in the area. A diagnosis can be made using ultrasound, with the hallmark finding of a “halo” phenomenon around the tendon [1, 6, 19, 71]. This increased accumulation of liquid around the tendon is most clearly visible in the transverse plane [5]. As climbers tend to have more liquid in their flexor tendon sheets after high stress on various ranges, no clear information can be given regarding the normal range [19, 67]. It is recommended to compare

the ultrasound finding of the injured finger to the same finger on the contralateral side [19, 67, 74]. A “halo” of more than 2 millimeters in a cross-sectional plane is strongly indicative of tenosynovitis [19] (Figs. 6.31 and 6.32). Color duplex

ultrasound can subsequently show the inflammation. In addition, an MRI examination can assist in the diagnosis but is usually not necessary (Figs. 6.33 and 6.34) [67].

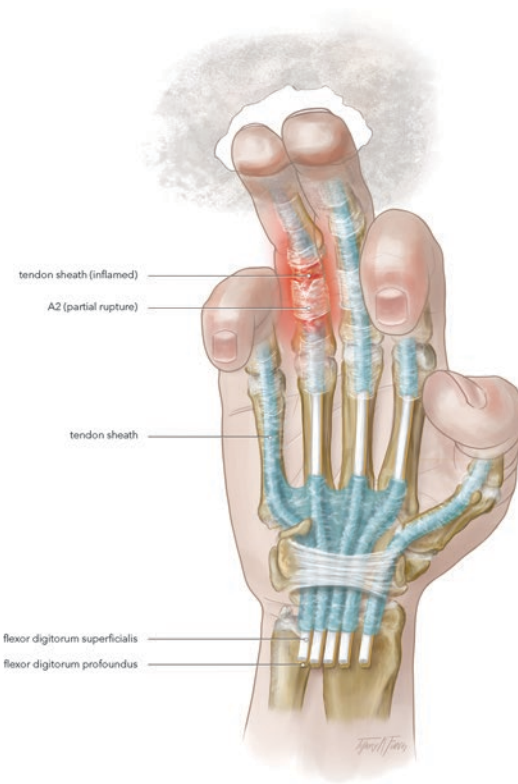


Fig. 6.30 Tenosynovitis after partial A2 tear (Pic by Tiffany Fung)

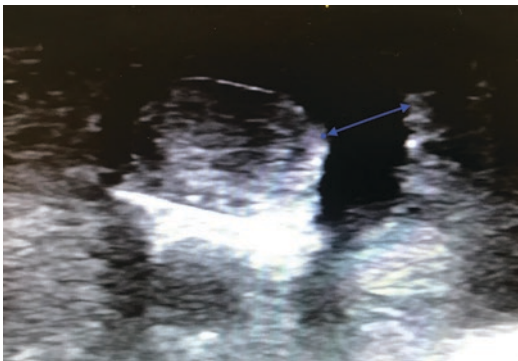


Fig. 6.31 “Halo” phenomenon in the ultrasound (the arrow shows the thickened tendon sheath)



Fig. 6.32 Normal finding for comparison

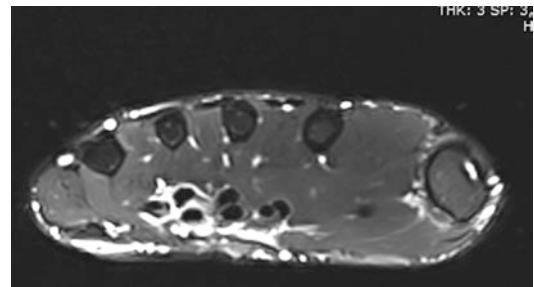


Fig. 6.33 Tenosynovitis of the palm in an MRI imaging



Fig. 6.34 Acupressure ring

6.3.4 Therapy

The initial therapy consists of anti-inflammatory medication, optional resting on a splint for several days, externals, brush massages, ice therapy and, in persisting conditions, local cortisone injections [19, 75]. These injections are not always avoidable, as chronic tenosynovitis can be stubborn. Surgical revision with tenosynovectomy is only required in rare cases. In mild cases, a stress reduction together with H-taping during sports will already decrease symptoms. If osteoarthrotic bone spurs are causing irritations to the tendons, removal of the bone spurs may be necessary. Alternative treatment options include radial shock wave therapy [76] or medicinal leech therapy [67].

For specific guidance in the treatment inflammatory flexor tenosynovitis, we developed a stage-related treatment regimen (Table 6.3) [67].

Stage 1 cases receive conservative therapy of self-massage with an acupressure ring (Fig. 6.34) or brush massages (with a toothbrush) to stimulate blood flow [67]. In addition, local overnight ointment dressings with Ichtolan® 20% (Ammonium bituminosulfonate, Ichtyol®, Hamburg, Germany) are applied for 10 days. Local ice therapy and nonsteroid anti-inflammatories can also be used [67]. The climbers are advised to either rest or to implement H-taping during climbing, depending on the extent of complaints or professional ability (e.g., during World Cup season, rest may not be possible). H-taping biomechanically deflects the angu-

lation of the tendon and thus decreases friction at the distal rim of the A2 pulley and the proximal rim of the A4 pulley [43].

If the tenosynovitis persists beyond 6 weeks, the condition is considered chronic (stage 2), and injections into the tendon sheath are recommended. Various substances such as corticosteroids (triamcinolone, dexamethasone), hyaluronic acid, platelet-rich plasma (PRP), or NSAIDs are used for injection therapy [77–81]. We mostly perform steroid or PRP injections. Although these injections have been reported to show beneficial effects [19], they are controversial since inadvertent intratendinous injection (instead of a peritendinous injection) of corticosteroids may cause tendon rupture [75, 82]. Therefore, correct injection technique is essential. Based on our own experiences, we favor a crystalloid corticosteroid (dexamethasone) in a prefabricated mixture with lidocaine (Supertendin®, Carinopharm, Elze, FRG), as this is the least viscous injection mixture and can be administered through a thin hypodermic (insulin) needle (27Gx0.5). Per injection, 0.1–0.2 mL is administered. The injection localization is defined as the point of the largest halo ring detected via ultrasound. The injection is performed in two areas. Following proper skin disinfection, the needle is inserted at a 60° angle pointing to the fingertip until it reaches the flexor tendons. The patient is then directed to flex the finger, causing the needle to move with the flexor tendon. The needle is retracted until the tendon can move freely, and only the tip of the needle remains in contact with the tendon, at

Table 6.3 Stage-related treatment regimen for tenosynovitis of the finger flexor tendons in climbers (Modified after Schöffl et al. 2019b)

Stage	Time frame since onset of pain	Therapy	Climbing rest
1	< 6 weeks	Conservative, icing, local therapy with acupressure ring	0–14 days, then stepwise climbing load increase with H-taping [18]
2	>6 weeks (or after failed conservative therapy >4 weeks)	Local infiltration with “Supertendin®” (corticosteroid) Reinjection after 7–10 days or radial shock wave therapy	No climbing or hand-related sports in between the injections and at least for 10 days after the second injection, then stepwise climbing load increase with H-taping [18]
3	Persistent pain >6 weeks after second injection	Medicinal leech therapy	14 days rest, then stepwise climbing load increase with H-taping [18]
4	Persistent pain >6 weeks after failed leech therapy	Surgical tenosynovectomy	6 weeks rest, then stepwise climbing load increase with H-taping [18]

which point 0.1 mL is injected (Fig. 6.35). In the second step, the needle is inserted further through the flexor tendons until it reaches the bone. This guarantees that the needle tip's open lumen is in the space between the deep surface of the flexor tendons and the bone, where the second part of the injection (0.1 mL) is administered (Fig. 6.36) [67]. We always aim to perform two injections 7–10 days apart and prescribe climbing rest for at least 10 days after the second injection.

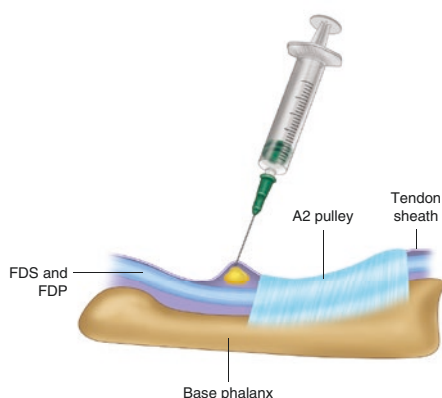


Fig. 6.35 Technique of steroid injection into the tendon sheath (Schöffl et al. 2019)

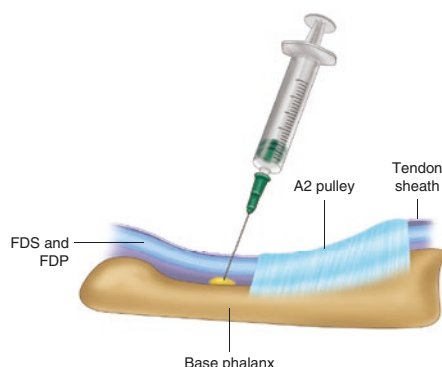


Fig. 6.36 Technique of steroid injection into the tendon sheath (Schöffl et al. 2019)

In an outcome analysis, we recently reported on 42 climbers with tenosynovitis, of which 31 climbers were free of pain (73.8%) after two steroid injections and a mean time frame of 21 days [67]. Only nine patients required another ongoing therapy. All athletes could restart their climbing, and all patients except one regained their pre-injury climbing level. No complication based on the injections occurred during the study [67].

Radial shock wave therapy can be used instead of injections at certain stages [73, 76]. Radial shock wave therapy has demonstrated effectiveness in the therapy of chronic, non-inflammatory tendinosis, based on an increased neovascularization [73]. Before administering radial shock wave therapy, the stage of the tenosynovitis needs to be evaluated carefully, as this technique is contraindicated in acute inflammatory tendonitis. Thus, looking at our strict regimen, it may be applied in stage 2 instead of injections.

In patients who do not respond to injection or radial shock wave therapy, we administer medicinal leech (*hirudo medicinalis*) therapy (stage 3) (Fig. 6.37). In this treatment, a medicinal leech is applied at the tenosynovitis site after needle puncture and left there until it lets go. Fourteen days of climbing rest are prescribed afterwards. Leeches act by withdrawing blood and injecting active substances through their saliva into the host tissue [83]. During this process, more than 100 substances are injected [83–85]. Active substances identified within leech saliva today include hirudin, factor Xa inhibitor, and hyaluronidase [83]. These substances have anticoagulant, thrombolytic, anti-inflammatory, and pain-relieving effects [83–85]. With these described effects as well as the positive effect on symptomatic osteoarthritis at the first carpometacarpal joint, we started using medicinal leeches as an alternative before surgery [86]. After positive preliminary results in our patients with tenosynovitis and capsulitis and in literature [83–85], we added this into our therapeutic regimen (stage 3).

A tenosynovectomy (stage 4 therapy) is only necessary in very rare conditions [19, 67]. In the aforementioned study, no tenosynovectomy was necessary in any patient [67].



Fig. 6.37 Medicinal leech therapy in a chronic tenosynovitis

- **History:** Overstrain, chronic, extensive crimping
- **Onset:** Acute, sometimes chronic
- **Clinical:** Point tenderness of the tendon sheath at the rim of the pulley, sometimes palpable swelling
- **Diagnostic:** Clinical finding, ultrasound, MRI optional
- **Therapy:** Conservative, in persistent pain >6 weeks local steroid injections (twice)
- **Outcome:** Good, surgical revision rarely necessary

6.3.5 Joint Capsule Damage and Collateral Ligament Injury

Pulling on one-finger pockets causes high stress upon the fingers' passive joint structures, the articular capsule, and the ligaments (Fig. 6.38). The climber also tries to reduce the strength



Fig. 6.38 Passive joint capsule stress in hanging onto a “mono” (Photo: E.Haase)

needed for pulling by jamming the finger in the pocket (Fig. 6.39). This can strain or rupture both the joint capsule or collateral ligaments.

An injury to the palmar plate typically occurs when the finger is jammed in a one- or two-finger pocket and while the climber tries to continue with the next move, resulting in a hyperextension force to the joint [19]. The diagnosis must include radiographic examination to exclude an associated avulsion fracture at any of the ligamentous attachments (Fig. 6.40). During a hyperextension injury of the PIP joint, the palmar plate can torn off the bone, sometimes with a little osseous fragment (Fig. 6.41). The collateral ligaments are assessed through ultrasound or fluoroscopy. Often, fractures can be visualized using ultrasound as well (Fig. 6.42). If joint capsuloligamentous injury is neglected, it can lead to flexion contracture or deviation deformity of



Fig. 6.39 Shear stress onto the fingers joint capsule in a one-finger pocket



Fig. 6.40 Osseous collateral ligament avulsion after jamming in a pocket

the finger (Fig. 6.43) with chronic decreased range of motion and dysfunction (Figs. 6.44 and 6.45). Chronic joint effusions and synovitis can also cause joint contractures.



Fig. 6.41 Fracture of the palmar plate caused by overextension and a stuck finger in a pocket

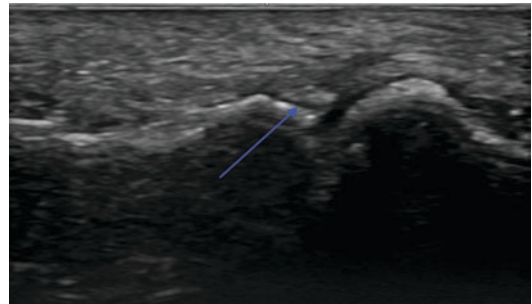


Fig. 6.42 Fracture detection in the ultrasound

6.3.5.1 Therapy

Depending on the stability of the joint, treatment is usually early functional range of motion exercises facilitated by buddy taping. A short period of immobilization may be necessary in the case of joint instability or an associated unstable fracture. Overall, the goal is as little immobilization as possible because prolonged immobilization leads to a



Fig. 6.43 In dislocation fused avulsion fracture of the palmar plate



Fig. 6.44 Contracture of multiple fingers in a long-time climber

higher rate of contracture. Only chronic instabilities of a high-grade require surgical treatment. It is important to remember that untreated joint capsule injuries frequently lead to these contractures, which



Fig. 6.45 Contracture of multiple fingers in a long-time climber



Fig. 6.46 Excessive inflammatory reaction after joint capsule injury. This case could be finally treated well with radiotherapy, but a persisting reduced range of motion resulted

are reported in a high number of older climbers (Figs. 6.44 and 6.45) [87–89]. In few cases, a prolonged inflammatory response can occur, which can be difficult to treat and will also lead to permanent articular damage (Figs. 6.46 and 6.47).

- **History:** Pulling on a one-finger pocket, finger jams
- **Onset:** Acute
- **Clinical:** Joint capsule swelling, painful reduced range of motion, joint instability, pain while pulling in the open grip position
- **Diagnostic:** Clinical finding, X-ray, ultrasound, optional fluoroscope
- **Therapy:** Early functional, rarely complete immobilization
- **Outcome:** Can lead to contracture

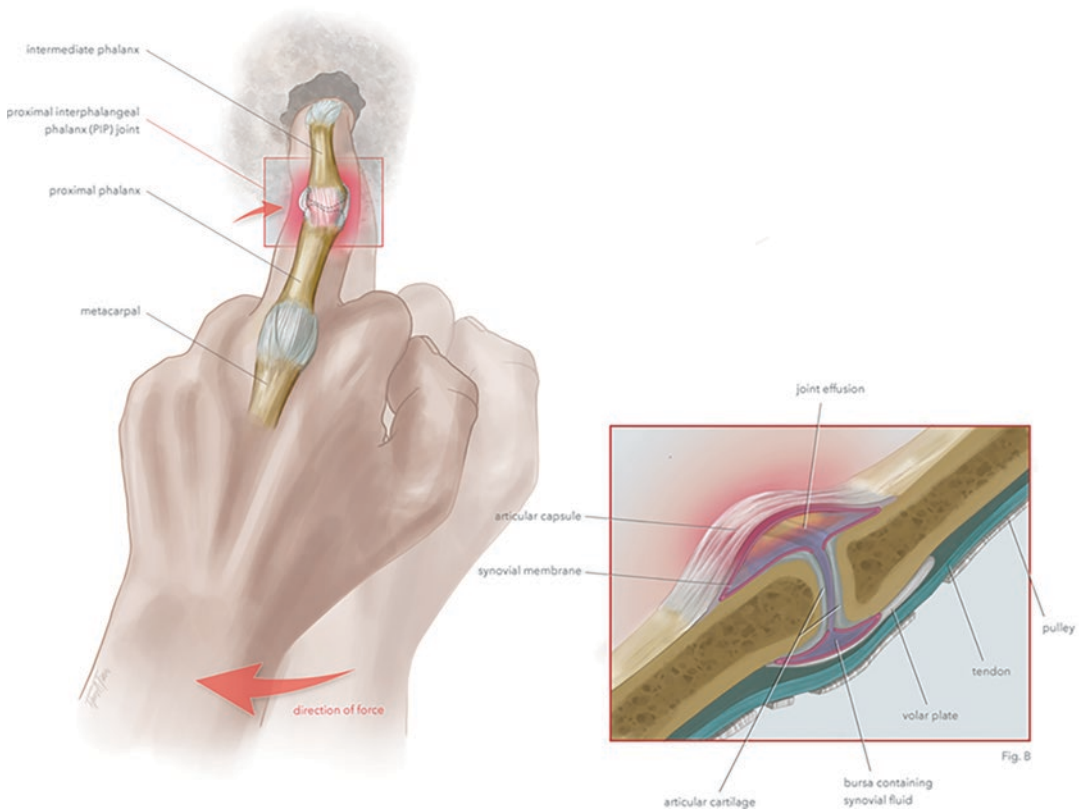


Fig. 6.47 Collateral ligament tear. (Pic by Tiffany Fung)

6.3.6 Chronic Capsulitis (Intra-Articular Synovitis)

High peak pressure within the finger IP joints during the crimp position causes the release of aggressive enzymes that can lead to a chronic inflammation of the joint capsule. If left untreated, chronic capsulitis is a precursor to chronic osteoarthritis (Figs. 6.48 and 6.49).

The term “capsulitis” is used to describe chronic synovitis of the finger (or toe) joints in climbing medicine. These can be caused by either repetitive stress or a single traumatic event with reactive effusion within the joint. The climber complains of early morning stiffness and swelling of the finger joints, a reduced range of motion, and pain, which often improves after activity (Fig. 6.50). The clinical examination shows a joint capsule effusion and typically dorsally sided pain under pressure and palpation. Overall, the increased thickness of the small finger joints, as

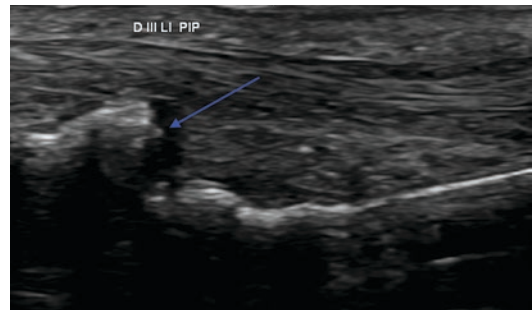


Fig. 6.48 Sonographic finding in a capsulitis

seen in many long-term climbers, is not a pathology per se (Fig. 6.51) but is rather thought to be a functional adaptation of the joint capsule and collateral ligaments to the stresses of climbing [87, 89]. About 40% of climbers have these thickenings of the finger middle and end joints [87, 89]; some cases are just soft tissue adaptations and others are chronic joint effusions [20]. Frequently, there is a correlation with extensive use of the

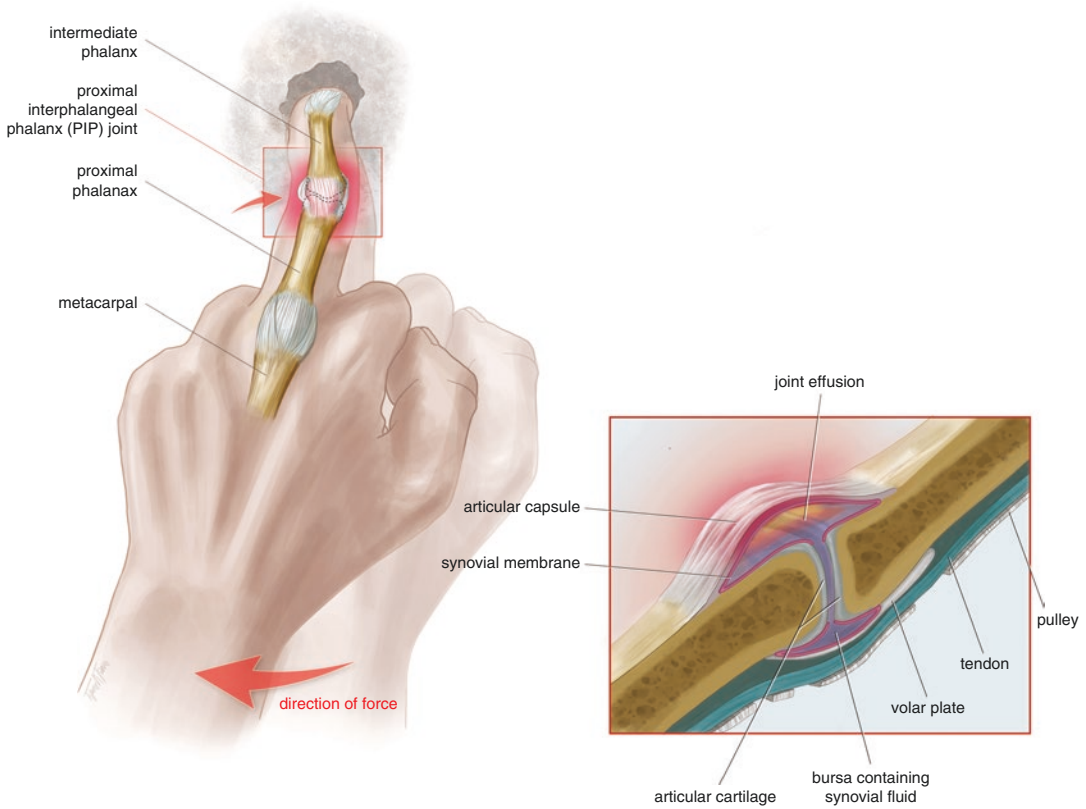


Fig. 6.49 Capsulitis following a collateral ligament tear (pic by Tiffany Fung)



Fig. 6.50 Finger middle joint swelling at the ring finger in capsulitis



Fig. 6.51 Stress adaptations of the joint capsule after more than 35 years of climbing

crimping position. In this position, the finger PIP joint is almost completely flexed, and the DIP joint is hyperextended. The stress on the cartilage is concentrated onto certain areas of the joint and is not evenly distributed over the whole articular surface. These high-pressure peaks cause micro-injuries and tears of the cartilage and a release of

aggressive enzymes into the joint. This can result in a reactive inflammatory reaction of the synovial membrane leading to hyperemia and effusion, which results in swollen joints in the clinical presentation and shows a joint space widening in X-ray examination (“X-ray sign 1”) [90]. Also the X-ray may show an inflammatory-edematous

synovial membrane in respect of a thickened soft tissue presentation (“X-ray sign 2”) [90]. This effusion itself stresses the synovia and can be the start of a vicious circle of overstrain, effusion, and synovitis, leading to a chronic capsulitis and, if untreated, to osteoarthritis and joint destruction [58, 90–93].

The chronic effusion is mostly visible on the dorsal aspect of the respective joint. It leads to a restricted range of motion, reduced fine-motor coordination, and dull pain. The dull pain is very characteristic and based on the chronic irritation of the joint capsule nerve endings.

Table 6.4 shows our therapeutic regimen. Immediate joint stress reduction by decreasing climbing volume and intensity is an essential first step in management. In addition, local anti-inflammatory therapy is performed using practices including but not limited to icing, movement therapy (e.g., TheraBand Hand Exerciser®, soft-ball, Qi-Gong balls, finger massage ring, or acupuncture ring), and traction therapy. In more chronic conditions, hand baths with a medical sulfur solution, ointment dressings (e.g., Ichtolan® 20% (Ammoniumbituminosulfonate, Ichtzol®, Hamburg, Germany), and exercising with therapy putty are used. If all of these fail, we perform fluoroscopy-guided intra-articular injections of steroid, hyaluronic acid, or platelet-rich plasma (PRP) (Fig. 6.52). Combinations of treatments are used frequently (e.g., initial treatment with an anti-inflammatory steroid injection followed by a PRP injection 1 week later). In cases of persistent symptoms, a radio-synoviorthesis with erbium shows good results [87]. Alternatively, medicinal leech therapy can be an effective treatment. We recommend an additional substitution with chondroitin and glucosamine.

- **History:** Extensive crimping, high stress load
- **Onset:** Chronic
- **Clinical:** Joint swelling, dull pain, pain while moving
- **Diagnostic:** Clinical finding, ultrasound
- **Therapy:** Difficult, often long-term, intra-articular injections, sometimes radiosynoviorthesis
- **Outcome:** Good, but may take a long time (1–2 years)

6.3.7 Osteoarthritis

The question of whether long-term high-level climbing causes osteoarthritis of the finger joints is widely discussed and also a topic of many scientific analyses. Studies report a positive correlation between climbing years and climbing level with osteoarthritic changes of the fingers [21, 94–97] (Figs. 6.53, 6.54, and 6.55). For further details regarding osteoarthritis in climbers, please refer to Chap. 13: Long-Term Effects of Intense Rock Climbing. The treatment options for osteoarthritis include movement therapy, anti-inflammatories (topical or oral), cryotherapy, intra-articular injections (e.g., steroid, hyaluronic acid, platelet-rich plasma, stem cells, autologous fat), and radiation therapy.

- **History:** Long-term intensive climbing, genetic predisposition.
- **Onset:** Chronic.
- **Clinical:** Joint swelling, decreased range of motion, pain.
- **Diagnostic:** Clinical finding, radiography.
- **Therapy:** Multiple options—conservative, injectable, radiation therapy, surgery.
- **Outcome:** The goal is to decrease symptoms and slow progression, with response to treatment and rates of degeneration varying widely.

6.3.8 Extensor Hood Syndrome

In athletes with a long history of climbing activity, progressive osteoarthritic changes of the small finger joints have been observed [88, 95, 98–100]. These changes can present as large bone spurs on both the flexor and extensor sides of the phalanges [65, 98] (Fig. 6.56). With repetitive use of the crimping position during climbing, these bone spurs can produce irritations to the extensor tendons [98] (Fig. 6.57). We reported on 13 rock climbers in a 3-year period complaining of dorsally sided pain of the proximal and/or the distal interphalangeal (PIP/DIP) joints [98]. Plain radiographs revealed dorsal bone spurs (osteophytes) on the PIP joint in all climbers and on the DIP joint in three climbers [98]. According to the Kellgren-

Table 6.4 Stage-related treatment regimen for capsulitis of the finger and toe joints in climbers

Stage	Time frame since onset of pain	Therapy	Climbing rest
1	<6 weeks	Conservative: icing, movement therapy, traction, hand baths with anti-inflammatory solution (e.g., medical sulfur solution), exercise therapy with therapy putty, acupressure ring, lymphatic drainage	• 0–14 days • Then stepwise climbing load increase in accordance to the clinical symptoms • Taping during climbing restart with a figure of eight tape
2	>6 weeks (or after failed conservative therapy >4 weeks)	2a: local infiltration with corticosteroid, reinjection after 7–10 days 2b: alternative local infiltration with PRP (platelet-rich plasma), mesenchymal stem cells (MSC), or autologous fat micrograft (if applicable combination of corticosteroid or PRP)	• 2a: no climbing or hand-related sports in between the injections and at least for 10 days after the second injection • 2b: no climbing or hand-related sports 48 h after the injection • Then stepwise climbing load increase • Taping during climbing restart with a figure of eight tape
3	Persistent pain and swelling/effusion >6 weeks after second injection	RSO (radiosynoviorrhesis)	• Immobilization of the respective joint for 48 h in a splint • Afterward light movement exercises; gradually restart of climbing activities after 6 weeks, with full load bearing after 10 weeks • Taping during climbing restart with a figure of eight tape
4	Persistent pain and swelling/effusion >6 month after first RSO	Second RSO if applicable in combination with simultaneous instillation of a corticoid or Medicinal leech therapy or Local radiation therapy (last resort: surgical synovectomy)	• Immobilization of the respective joint for 48 h in a splint • 6 weeks rest, then stepwise climbing load increase with full load bearing after 10 weeks • Taping during climbing restart with a figure of eight tape • Personalized follow-up regime in case of medicinal leech therapy, radiation, or synovectomy (prolonged immobilization until assured wound healing)



Fig. 6.52 Injection under fluoroscope monitoring



Fig. 6.53 Osteoarthritis in many finger joints in an 60-year-old climber who still climbs at a UIAA level 9 (YDS 5.12.c, Fr 7b+)

Lawrence scale [22], the radiographs (found bilaterally in seven cases) revealed 5 grade 2, 12 grade 3, and 3 grade 4 osteoarthritis. Each of these dorsal bone spurs was causing irritation of the extensor mechanism, resulting in fluid accumulation and tendonitis. In two cases, the dorsal osteophyte was fractured at the time of presentation [98].

Treatment for extensor hood mechanism is primarily conservative with anti-inflammatory ointment dressings, anti-inflammatory hand baths, or local steroid injections. In rare cases, however, a surgical excision of the dorsal-sided bone spurs is necessary to alleviate the extensor tendon irritation [98].



Fig. 6.54 Osteoarthritis in many finger joints in an 60-year-old climber who still climbs at a UIAA level 9 (YDS 5.12.c, Fr 7b+)

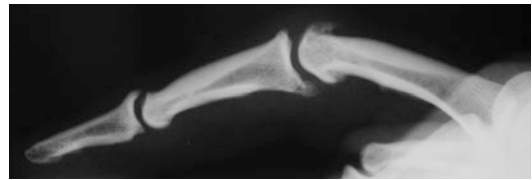


Fig. 6.55 Radiographic finding of osteoarthritis in a long-time climber (over 35 years of climbing). Surprisingly, the climber is almost pain-free!

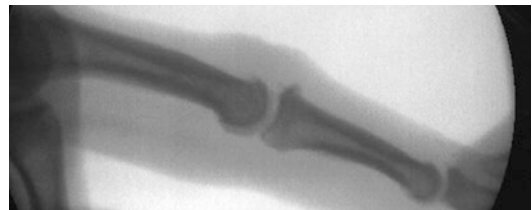


Fig. 6.56 Extensor hood syndrome in fluoroscope imaging



Fig. 6.57 Extensor hood syndrome

- **History:** Many years of climbing, with dorsal osteophytosis of the IP joints and gradual onset of dorsal-sided finger pain
- **Onset:** Chronic
- **Clinical:** Dorsally sided pain over the fingers' middle or end joint
- **Diagnostic:** Clinical finding, radiographic, ultrasound
- **Therapy:** Mostly conservative
- **Outcome:** Good with conservative treatment, though sometimes surgical treatment is necessary

6.3.9 Flexor Tendon Strains and Tears

Directly injured flexor tendons were observed in relatively few cases, most often caused by a sudden stress on a hand or finger in a hanging posi-



Fig. 6.58 Complete rupture of the deep flexor tendon after a one-finger pocket move

tion (e.g., the foot slipping off a foothold) [19, 74] (Figs. 6.58 and 6.59). Tendon strains occur most frequently, whereas direct tears, complete or partial, are rare. Patients with flexor tendon strains present with pain running along the flexor tendon [58]. This pain increases in the open finger grip position and can sometimes be absent in a crimping position. Often, a strain of the lumbrical muscle is also present, and lumbrical shift sign will be positive. Physical examination should include isolated testing of the flexor digitorum profundus and superficialis, with the deep flexor tendon more commonly injured. The deep flexor tendons (FDP) are mostly affected. For further diagnoses, ultrasound and MRI are utilized. In flexor tendon strains, the recovery can be prolonged, and the reoccurrence rate is high. Therapy is conservative combined with therapeutic ultrasound, radial shock wave ther-



Fig. 6.59 Complete rupture of the deep flexor tendon after a one-finger pocket move

apy, and light eccentric training in a later stage of the rehabilitation. In rare cases, a partial tear of the tendon occurs, which can lead to tendon nodules and triggering [101, 102].

Open tendon injuries are common findings in general trauma and orthopedic patients and require primary surgical treatment. Closed lesions are more difficult to detect. A differentiation between complete or partial rupture must be made. A lesion of less than 60% of the tendons diameter should be treated conservatively [103]. Multiple biomechanical in vivo and ex vivo studies have demonstrated that conservatively treated partial tendon ruptures have a higher tear resistance ($p < 0.05$) compared to those which are surgically treated [103]. Chronic degenerative tendon ruptures such as those seen in rheumatoid arthritis patients must be distinguished from these traumatic ruptures. In rock climbers, we see a combined entity, with a chronic tendinosis and degeneration and, finally, an acute tear. High intensive stress combined with chronic tenosynovitis and tendinosis can lead to a rupture of the tendon, even after minor mechanical stress [19]. Additionally, the poor blood supply of the ten-

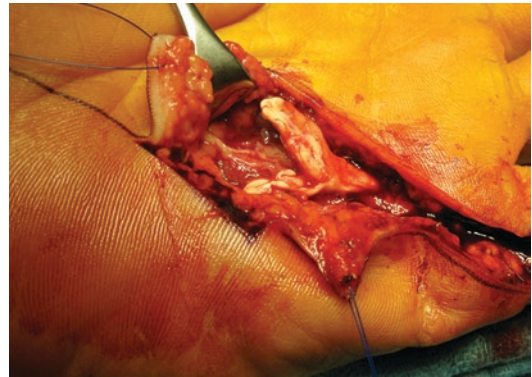


Fig. 6.60 Degenerative spontaneous rupture of both flexor tendons of the small finger while climbing

don, microtrauma, and microscopic structural damage can lead to a modest prognosis.

These complete tendon tears are surprisingly rare and require surgical treatment (Fig. 6.60). Usually, these degenerative attrition tendon ruptures leave the tendon in poor condition for repair, so other reconstructive surgical techniques are required, such as arthrodesis or tendon transfer. If the tear is at the level of the distal interphalangeal joint, flexor tendon reconstruction using free tendon grafting or DIP joint arthrodesis are the best options [19, 58, 104].

Flexor tendon closed avulsion injuries occur at the insertion of the FDP tendon to the distal phalanx. This injury is seen most frequently on the fourth finger, where the FDP-tendon is embedded in between the double-sided lumbrical tendons as shown by Manske and Lesker [105] in cadaver dissection. We have verified that this injury in rock climbers occurs in the setting of chronic degenerative damage to the tendon [104, 106].

These injuries often result in poor postoperative outcomes [19]. Four factors determine the prognosis of avulsion injuries: the extent of the retraction of the tendon, the remaining blood supply, the time interval between trauma and surgery, and the presence and size of an osseous fragment [107]. Recently, Schweizer et al. [108] reported on closed disruption of a single flexor digitorum superficialis tendon in three cases, two in climbing, one in judo. With conservative and functional treatment, all three patients regained

normal function and return to sport. The authors concluded that the disruption of one FDS tendon slip is a differential diagnosis for a popping incident in the finger, which is typically associated with a closed flexor tendon pulley injury [108].

In rare cases, osteophyte irritation of the flexor tendons can occur and may require surgical removal to relieve pain and prevent future attrition rupture of the tendon [65, 74] (Fig. 6.61).

- **History:** Mostly degenerative, due to tendinosis. In rare cases, acute blunt disruption.
- **Onset:** Acute.
- **Clinical:** Loss of function, pain in a hanging finger position. Frequently in combination with lumbrical muscle strain or tear.
- **Diagnostic:** Clinical findings, ultrasound, MRI optional.



Fig. 6.61 Flexor tendon irritation based on a palmar-sided bone spur, which later needed surgical removal. The climber later regained international top climbing level

- **Therapy:** Conservative mostly, surgical in complete tears.
- **Outcome:** Recovery can be prolonged. Sometimes a joint fusion or tendon reconstruction is necessary.

6.3.10 Ganglion Cysts

Flexor tendon sheath and pulley ganglion cysts are frequent in rock climbers [74, 109, 110]. They are caused by high local pressure onto the flexor tendon sheath from sharp edges of finger holds, leading to microinjuries and tears which eventually form a ganglion cyst. However, flexor tendon sheath and pulley ganglia are not exclusive to climbing [111]. In climbers, they mostly occur on the palmar side but sometimes also on the extensor side, often in combination with a joint effusion (Figs. 6.62 and 6.63). Clinically, they are easy to detect; otherwise, ultrasonography can confirm the diagnosis (Figs. 6.64 and 6.65) [6, 19, 70, 112]. The treatment is conservative; sometimes, the cysts dissolve by themselves through direct pressure applied either while climbing or from gentle self-massage. Sometimes, a puncture and local steroid instillation are necessary. If these conservative options fail, surgical excision can be performed. An open surgery technique is used in order to visualize and protect the digital nerves during the ganglionectomy procedure.



Fig. 6.62 Extensor sided ganglion

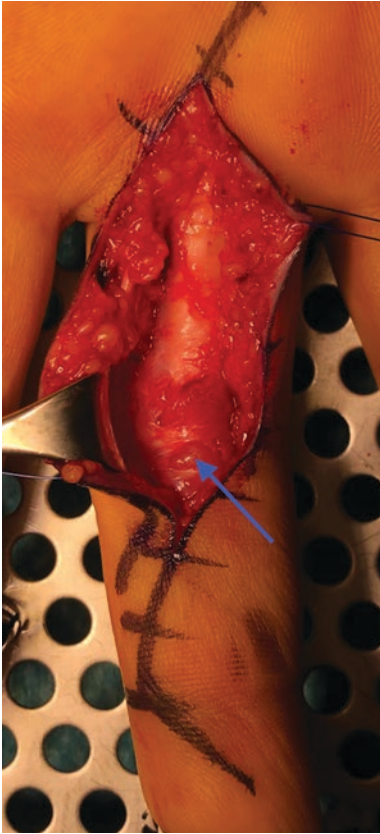


Fig. 6.63 Pulley ganglion as a random finding during surgery

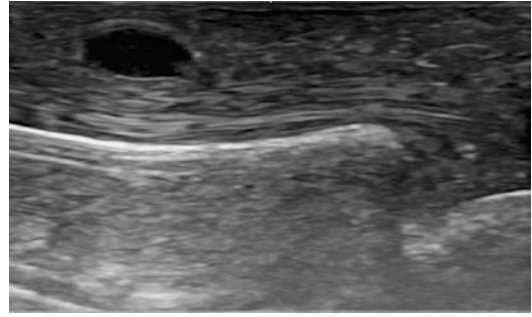


Fig. 6.65 A2 RB pulley ganglion in a longitudinal ultrasound plane

- **Therapy:** Conservative, sometimes steroid instillation
- **Outcome:** Good

6.3.11 Epiphyseal Growth Plate Fractures in Adolescent Climbers

Epiphyseal fracture of the finger middle phalanx base is an injury specific to high-level adolescent sport climbers [113]. Since the first reported case in 1997, there have been more than 100 of these fractures reported in literature [113–117], and recent reports show an increase in incidence [110, 118, 119]. Presently, they are reported to be the most frequent climbing-specific injury in adolescent climbers [113]. Most recently they have also been related to an increase in speed climbing training [120]. With the inclusion of climbing in the Olympics, a continuously increasing number of these injuries are to be expected, as overall training time and intensity will increase and the age of high level climbers will continue to decrease [121]. Patients are typically between 13 and 15 years old and in the phase of the pubertal growth spurt. The fractures are mostly “transient” fractures, meaning fractures during the phase of growth plate closure, where the palmar part of the growth plate is already fused and the dorsal part is still widely open [122] (Fig. 6.66) The middle or ring fingers are most commonly affected, and these fractures are believed to be caused from repetitive stress and microtrauma rather than from a single acute event [113].

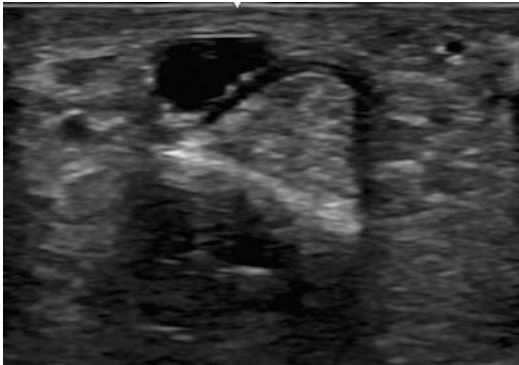


Fig. 6.64 A2 pulley ganglion in a transverse ultrasound plane

- **History:** Microtrauma
- **Onset:** Slow
- **Clinical:** Pressure tenderness, palpable nodule
- **Diagnostic:** Clinical finding, ultrasound



Fig. 6.66 Epiphyseal growth plate fracture as a transient fracture (age: 13 years)

Nevertheless, in a higher number of cases, the patients refer to a final, evolving event, e.g., one hard campus board workout or one exceptionally intense bouldering session.

These fractures are caused by the high pressure onto the dorsal aspect of the growth plate in a crimping position [113]. In addition, the pulling forces of the extensor tendon central slip is at the dorsal growth plate [115, 122, 123]. Bartschi et al. [124] also report a translatory shift of the fingers' middle phalanx base on the proximal phalanx head as a possible additional cause. The pulling force of the flexor tendons shifts the dorsal cortex in a crimping position dorsally, while the pressure from the palmar side onto the fingertips during crimping also pushes the cortex dorsally. Thus, two forces are applied to the dorsal growth plate fragment from two sides, shifting it dorsally.

We found that out of 16 youth climbers with finger pain, 14 revealed an epiphyseal fracture on magnetic resonance imaging and one climber showed bilateral fractures [110]. Of 20 injured young climbers up to the age of 14, 14 (70%) had an epiphyseal fracture [110]. Thus, an epiphyseal growth plate fracture is by far the most frequent injury in young climbers [125], with the overall incidence still increasing [95].

6.3.11.1 Diagnosis

The growth plates of the finger bones are not closed until the age of 17–19 years. Growth plates are the weakest structure of the finger in adolescents and are most susceptible to injuries [115]. As soon as pain without an obvious trauma in the PIP joints is apparent in an adolescent, the crimp grip position should not be allowed at all until the pain disappears or the growth plates have closed at the age of 17–19 years. In the clinical presentation, the climbers report dorsally sided pain at the PIP joint, mostly at the base of the middle phalanx. There is pressure tenderness at that area, sometimes accompanied by minor swelling and a palpable protruding bone fragment (Fig. 6.67).

After clinical suspicion, a standard radiograph can be performed to diagnose a fracture (Fig. 6.66). It is important to note that the mostly obliquely oriented fracture lines are obscured and may not be visible in conventional radiographs. If the plain radiographs are negative and there is a high index of suspicion for fracture, CT scan and/or MRI should be performed to rule out an occult fracture [123] (Fig. 6.68). A standardized diagnostic-therapeutic algorithm is currently published [126]. We recommend an MRI in all adolescent climbers with persistent finger pain for 2 weeks after sufficient rest and in the absence of acute finger trauma.

The fracture pattern is usually Salter-Harris type III (81%) [113, 114] on mostly the middle finger but sometimes the ring finger [63, 113, 114, 125]. Sonographic detection of these injuries is far more difficult. While many of them are detectable, some are not; thus, a normal ultrasound does not exclude an epiphyseal fracture. Nevertheless, as described in prior chapters, all

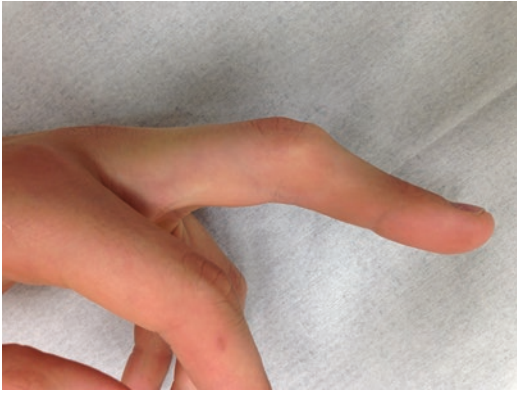


Fig. 6.67 Clinical finding in an epiphyseal growth plate fracture; note the protruding dorsal bone



Fig. 6.68 Epiphyseal growth plate fracture in the MRI. The normal radiographs could not detect a fracture (age: 13 years)! MRI diagnostics are mandatory in clinical suspicion if the normal radiographs are negative

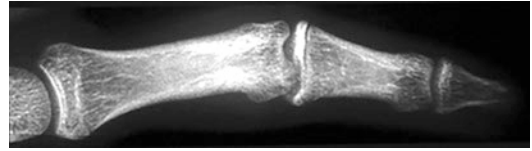


Fig. 6.69 Malunion of a growth plate fracture. No therapy was performed

finger injuries receive a diagnostic ultrasound evaluation in our clinic. If growth plate injuries are not diagnosed and neglected, they lead to permanent PIP joint osteoarthritis, often with stiffness, pain, and deformities (Fig. 6.69).

6.3.11.2 Therapy

Non-displaced fractures receive conservative therapy, including complete rest from climbing and gentle finger range of motion exercises until fracture healing, which usually occurs within 2–4 months [115]. Immobilization is usually not necessary, but the climber must cease all finger-stressing exercises until fracture healing has been confirmed. This also includes isolated hang board training with the neighboring fingers, as the extensor tendons are interconnected at the level of the metacarpophalangeal joint. Due to this connection, a training-related stress onto the ring finger also leads to a stress to the extensor tendons of the middle finger. This results in increased stress of the middle finger's growth plate, even if the middle finger is not directly involved in gripping during the exercise. A stepwise return to climb regimen was recently published by Meyers et al. [127]. Results of conservative management are excellent in patients who present with acute, non-displaced fractures. Patients presenting with chronic or displaced fractures may heal with conservative treatment, but some have incongruity of the middle phalanx base at the fracture site. Patients who continue to climb without allowing their fractures to heal are likely to develop osteonecrosis of the epiphyseal fracture fragment, with resultant permanent angular deformity and stiffness of the affected joint. Acute fractures with significant displacement are treated with closed



Fig. 6.70 Therapy-resistant sclerosed epiphyseal growth plate fracture (age; 15 years, female)



Fig. 6.71 Surgical spot drilling in patient of Fig. 6.70

reduction $\pm$ percutaneous k-wire fixation. Rarely, an open reduction is considered if closed treatment fails; however, risk of an open procedure to the growth plate must be considered. In the case of mild displacement of the fracture, we still treat conservatively but immobilize for 2–3 weeks. In cases with a delayed osseous healing, we perform a CT scan to evaluate for sclerosis at the fracture margins, a sign of established nonunion, which will prevent bone healing without surgical manipulation (Fig. 6.70). If a CT scan shows a displaced fracture and sclerosis, we treat the injury with surgical epiphysiodesis via spot drilling [122] (Figs. 6.71 and 6.72). Recently a therapeutic algorithm was established and evaluated [128].

6.3.11.3 Results

In our series of 16 youth climbers with finger epiphyseal fractures, we found that nine (56%)

showed complete healing, five (31%) had incongruence of the joint space, and two (13%) cases resulted in a deformity. In another study, we found that 70% of the youth climbers with epiphyseal fractures had good outcomes, whereas 23% were still pending and one case (4.5%) had permanent damage [113]. Lastly, we reported that intensive finger training (campus boarding, a previously known risk factor for epiphyseal fractures) during adolescence can lead to early-onset osteoarthritis of the hand up to a decade later [95]. We also described good outcome of patients treated with surgical spot drilling epiphysiodesis [122]. To date, we have performed these spot drillings in about ten cases, all with good results and complete fusion and fracture healing within 8–10 weeks.

The most important intervention factors for prevention of epiphyseal fractures are sports

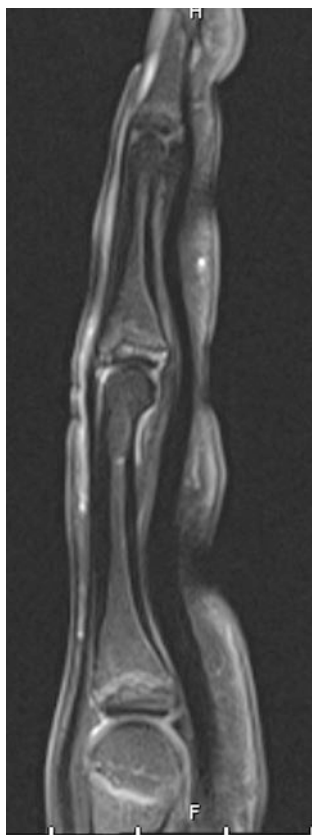


Fig. 6.72 MRI with complete fusion after 4 months

medical education and close sports medical surveillance of competitive adolescent climbers, with the avoidance of certain high-stress training exercises in this age group (e.g., campus board, training board with extra weight). The use of the campus board was shown in several studies to be a significant risk factor for the development of an early onset of finger joint osteoarthritis in adolescent high-level climbers [95]. Some eager trainers argue against this, stating that the junior climbers are also crimping during climbing, which is also provoking the injury. Nevertheless, we think that the extra stress of an additional campus board training or hang board training with extra weights certainly adds to the overall training load and may not even be necessary in this age group. Even though the pubertal growth spurt is considered the high-risk phase for these fatigue fractures, we have seen them in younger athletes without

any signs of puberty onset (Tanner stage 0) and in athletes over the age of 18. Close training monitoring with analysis of the overall training load and avoidance of overly intensive load peaks is important. Additionally, sports medical supervision and annual clinical examination of the hands should be performed in youth climbers. Overall, finger pain in an adolescent after or during climbing that does not abate after 1 week of rest should be managed as an epiphyseal fracture until proven otherwise [113].

In conclusion, early detection of a growth plate fracture is critical to ensure high likelihood of healing without long-term consequences. Thus, educating sports medical doctors, trainers, athletes, and parents is crucial in order to instill acute clinical awareness for this injury in youth climbers with finger pain at the PIP joint level.

- **History:** Chronic, mostly crimping position
- **Onset:** Slow, sometimes acute initiating event
- **Clinical:** Dorsal-sided PIP joint tenderness
- **Diagnostic:** Clinical finding, MRI
- **Therapy:** Mostly conservative, sometimes surgical spot drilling epiphysiodesis
- **Outcome:** Good if treated, very bad if neglected (permanent damage)

6.3.12 Lumbrical Shift Syndrome and Lumbrical Muscle Tears

Injuries to the lumbrical muscles have been reported to be rare climbing-specific injuries by Schweizer et al. [129]. In recent years, we have seen an increasing number of these injuries in our patients as well as in international, top-level climbers [109, 121, 130].

The pathomechanics of lumbrical muscle injury are related to the “quadrige effect” (Fig. 6.73), which describes a shear stress on the bipennate third or fourth lumbrical muscles to the ring and small fingers, respectively. These two lumbricals frequently have a dual origin from adjacent flexor digitorum profundus tendons, so shear stress injury to the muscle can occur [129–131] from gripping positions in which one or two fingers are extended while the neighboring fin-

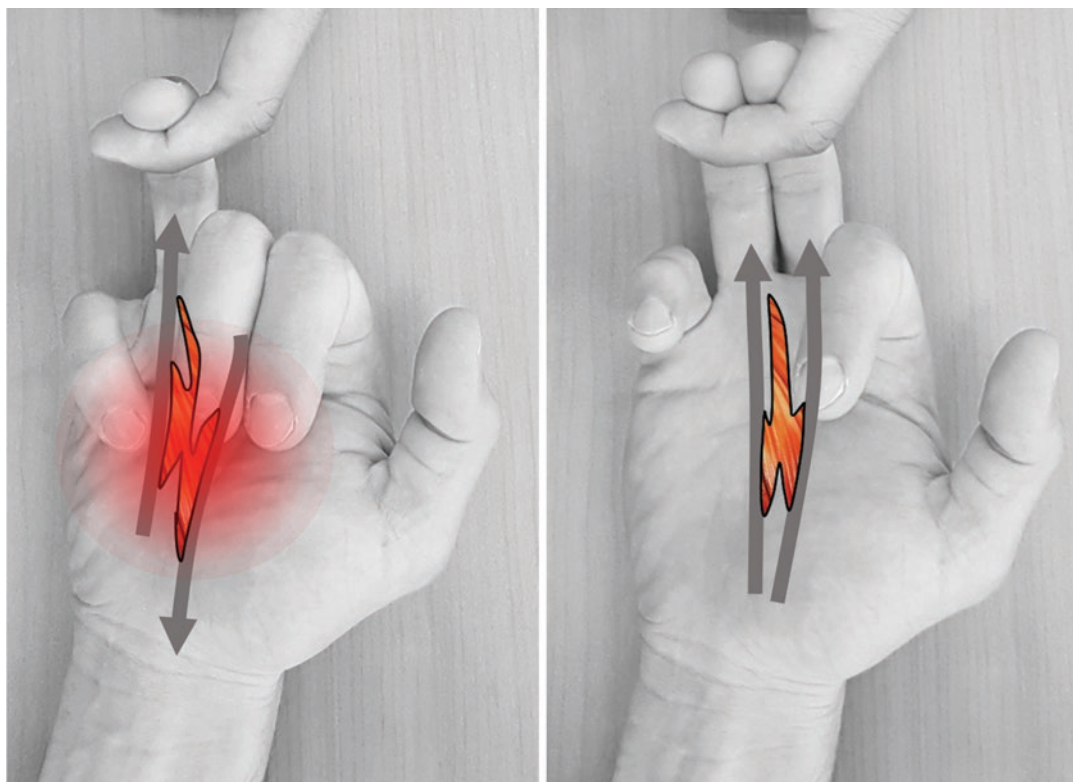


Fig. 6.73 Lumbrical shift test

gers are flexed by force [129, 131]. Typically, this occurs in one- or two-finger pockets.

This finger position increases the maximum strength by up to 50% but causes a shift of the FDP tendons and the common muscle body of the various fingers against each other, leading to muscle strains or partial tears [130] (Fig. 6.74). Injuries typically occur within the third or fourth lumbrical muscles because these are the only ones that have a bipennate origin [130, 132–134].

6.3.12.1 Diagnosis

The diagnosis is made using the lumbrical stress test (Fig. 6.1). This test is positive when pain is elicited while loading the injured finger in extension, with the adjacent fingers flexed. The pain is at the palm, sometimes radiating into the forearm along the course of the FDP tendon of the respective finger (Fig. 6.75). If the climber pulls with all fingers in extension, there is no pain. Also, crimping is usually pain-free. A detailed

overview shows the three different grades of injuries (Fig. 6.5). Ultrasound shows an edema of the lumbrical muscle (grade II injury) or even a hematoma (grade III injury). In cases of hematoma (grade III injury), an MRI should be performed (Figs. 6.76 and 6.77).

6.3.12.2 Therapy and Outcome

Therapy consists of symptomatic treatment, taping, and carefully stretching of the muscle [129] and follows our guidelines (Fig. 6.78) [135]. Initially, a rest from climbing and careful movement exercises are recommended. Immobilization is only required in high-grade injuries. Immobilization of the injured lumbrical muscle is initially performed through buddy taping and later through specific lumbrical taping (see Chap. 22 “Taping”). It is important to start with stretching exercises early, which is done in the same way that the injury was provoked but with much less load [27]. In addition, radial shock wave therapy has been shown to provide good pain



Fig. 6.74 Typical two-finger pocket position with lumbrical stress. (Pic by Tiffany Fung)

relief [74]. Healing time is often prolonged with a high reoccurrence rate if exposed to similar stress. Lumbrical injuries are often correlated to flexor tendons strains.

We reviewed data from 60 consecutive patients with a positive lumbrical stress test, including clinical examination ($n = 60/60$), ultrasound ($n = 60/60$), magnetic resonance imaging ($n = 12/60$), and outcome ($n = 60/60$) [130]. Lumbrical muscle tears were graded as mentioned above (grade I–III injuries) [130]. The therapy consisted of adapted functional therapy [130]. Thirty percent of patients had grade I injuries (micro-trauma), 53% had grade II injuries (muscle fibre disruption), and 16% had grade III injuries (musculotendinous disruption) [130]. All patients had an uncomplicated outcome with complete recovery and unaffected return to climbing [130]. The healing period in grade III



Fig. 6.75 Lumbrical muscle tear with clear swelling of the injured finger

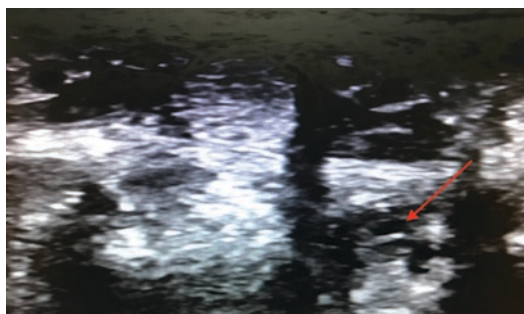


Fig. 6.76 Ultrasound of the lumbrical rupture in the same patient

injuries was significantly longer than in the other two groups ($p < 0.001$).

- **History:** One or two-finger pocket
- **Onset:** Acute
- **Clinical:** Positive lumbrical stress test
- **Diagnostic:** Clinical finding, ultrasound, MRI
- **Therapy:** Conservative
- **Outcome:** Often prolonged healing time, high rate of reoccurrence

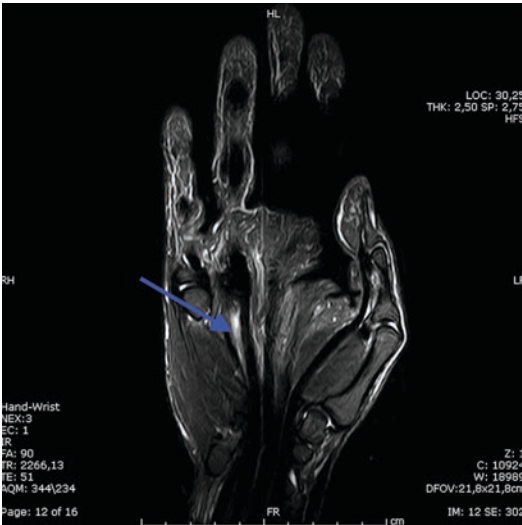


Fig. 6.77 Lumbrical rupture in the MRI

6.3.13 Cellulitis and Other Soft Tissue Infections

Climbing-related minor cuts of the skin can cause bacterial infections of the skin (cellulitis), which can spread to the deep soft tissues of the finger and the flexor tendon sheath (suppurative tenosynovitis) (Fig. 6.79). The classic clinical findings of suppurative tenosynovitis are Kanavel’s cardinal signs (fusiform finger swelling, tenderness of the flexor sheath, pain with passive extension, and slight flexed position of the finger). The ultrasound can help in the detection of an edema or pus in the tendon sheath. In the early stages (24–72 h), suppurative tenosynovitis can sometimes be treated conservatively, with splinting, elevation, and oral or IV antibiotics. However, after this initial period, urgent surgical incision

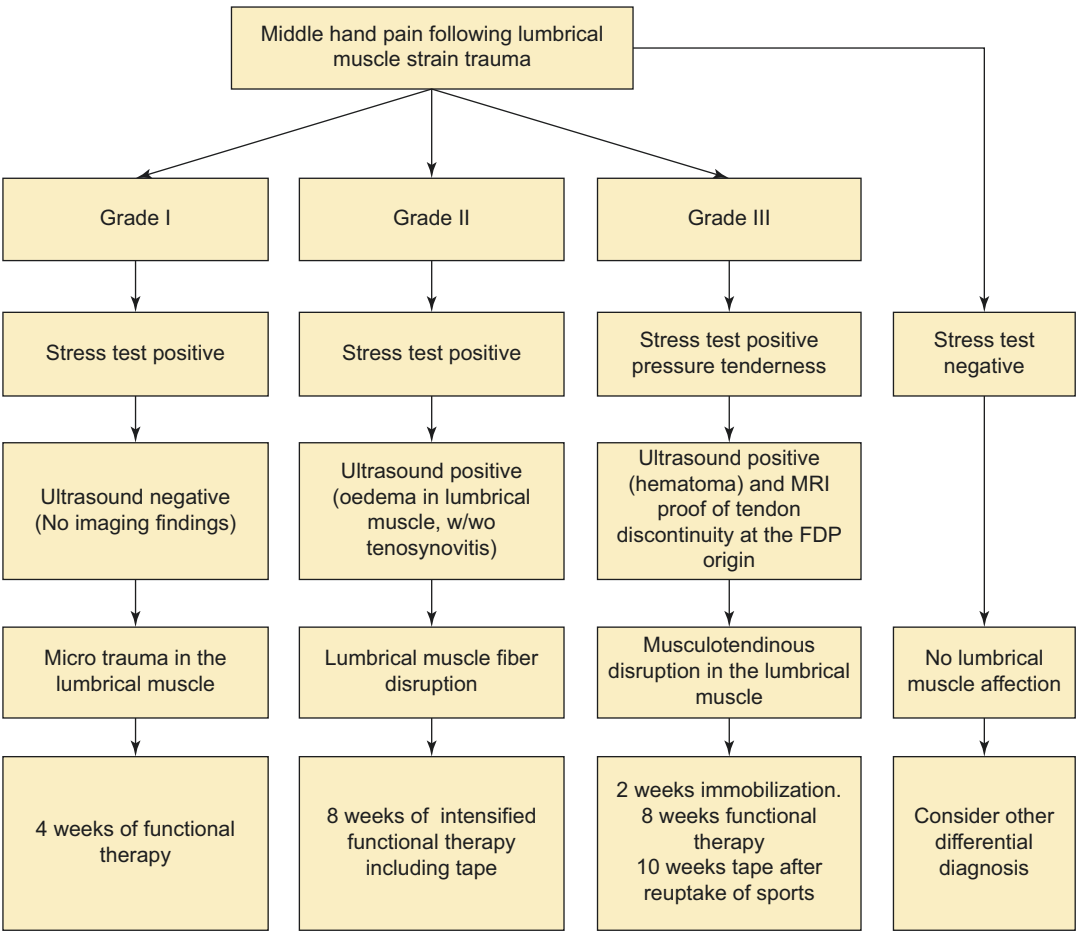


Fig. 6.78 Lumbrical rupture in the MRI



Fig. 6.79 Skin cuts are frequent

and drainage +/- debridement are necessary to clear the infection before necrosis of the flexor tendons occurs (Figs. 6.80 and 6.81).

- **History:** Small skin breaks (cuts)
- **Onset:** Over a few hours to days
- **Clinical:** Kanavel's cardinal signs
- **Diagnostic:** Clinical findings, ultrasound
- **Therapy:** Initially conservative, otherwise surgical
- **Outcome:** Dependent on early diagnosis and treatment

6.3.14 Irritations of Finger Nerves

While nerve compression syndromes in climbers (e.g., carpal tunnel syndrome, supinator syndrome) are described and explored in several studies [136, 137], finger digital nerve irritation is a less commonly discussed problem and occurs almost exclusively in climbers, particularly when climbing on sharp finger pockets or jamming in finger cracks. If the climber uses one- or two-



Fig. 6.80 Skin injuries can easily lead to infections (cellulitis)



Fig. 6.81 Nailbed infection after a minor injury

finger pockets, the fingertip is often jammed into the pocket (Fig. 6.82) [58], resulting in localized pressure on one or both digital nerves at the level of the pocket edge. If this pressure is high-load, prolonged, or repeated, it can result in a neuropraxic injury to the digital nerve, which causes a temporary nerve conduction block (i.e., mild acute compression neuropathy), with resultant numbness, paresthesias, or neuropathic pain in the corresponding finger digital nerve territory. It is difficult to predict how long the recovery time of the respective nerve will be, but most resolve within a few days to weeks.



Fig. 6.82 Jamming in pockets can lead to finger nerve damage

Therapy is conservative, consisting of regular finger massage, range of motion, and nerve gliding exercises as well as explanation of the cause of the problem to reassure the climber and help prevent further injury.

- **History:** High localized pressure onto the fingers digital nerve (s)
- **Onset:** Acute
- **Clinical:** Numb fingertip
- **Diagnostic:** Clinical finding
- **Therapy:** Conservative
- **Outcome:** Excellent

6.3.15 Dupuytren Contracture

Historically, this is a disease that usually occurs in men between the age of 40 and 60 and is caused by genetic predisposition, with other risk factors including diabetes, hypothyroidism, epilepsy, and chronic microtraumas associated with hard labor [138–140]. However, this condition is now on the rise in climbers, also those younger than 40. In a study published by Logan et al. in 2005, an increased prevalence of 20% was described for Dupuytren's disease in a relatively young rock climber sample (mean age 54 years) [141]. We have even seen several cases of early Dupuytren's contracture in climbers under 18 [19]. This condition is so rare among younger



Fig. 6.83 Dupuytren's contracture in a climber

people that a number of young climbers who have suffered from it have actually been misdiagnosed with a malignant tumor [19]. In climbers, it is most likely caused by the repetitive stress and microtrauma to the palmar aponeurosis, which triggers the disease [19, 141]. Pervulesko et al. developed a self-diagnostic tool for rock climbers which was also used to evaluate Dupuytren's contracture frequency in climbers [142].

The therapy is prescribed according to severity and is mostly conservative. Most climbers only present with early stages, which rarely progress extensively (Fig. 6.83). In higher-grade stages (grade 3–4), various treatment techniques are available, including needle fasciotomy, Xiaflex® collagenase injection, radiation therapy, or selective fasciectomy surgery.

- **History:** Eecurrent mechanical stress onto the palmar aponeurosis
- **Onset:** Slow
- **Clinical:** Palpable and visible nodule
- **Diagnostic:** Clinical finding
- **Therapy:** Mostly conservative
- **Outcome:** Stage-related

6.3.16 Rope-Tangling Injuries, Avulsion Amputations

Rope-tangling injuries are an established pathology in climbing accidents and were first reported in 2003 [143], followed by subsequent reports

which had different causative mechanisms [144, 145]. There are three major mechanisms involved with rope-entanglement or rope-catching injuries:

1. A sport climber takes a minor fall, just above the protection. While falling, he or she may grab the ascending rope, mostly as a reflex borne out by fear. This grabbing action is unnecessary, as “catching a fall” is the responsibility of the belayer at the other end of the rope. However, by instinctively grabbing the rope close to the protection point, the following injuries are possible. The climber may (1) catch the fall by him- or herself and get severe skin and soft tissue injuries in the palm, (2) get tangled with the belay point, (3) unwillingly unclip the belay device which may create the potential for more serious injuries, or (4) tangle the rope around his or her hand or fingers. All of these causative mechanisms have been reported [144].
2. A climber takes a long fall and unintentionally grabs the rope that comes from the harness. If the fall is long, more loose rope is involved. This longer rope can tangle or loop around the climber’s hand or fingers, and, once the rope tightens at the moment of the fall impact, the resulting tension causes avulsion amputations of one or more digits, fractures, or severe contusions/bruises [144, 145].
3. A climber takes a fall on a steep or overhanging route. As he or she is dangling by the rope beneath the last point of protection, the standard technique in sport climbing to get back onto the wall is to perform one or two pull-ups on the rope and then let go. With each pull-up, the belayer takes in the excess rope or runs backward to reestablish rope tension. Thus, the climber can slowly make his way up the rope. However, we have one case study of a very experienced high-level climber who entangled his hand with the rope while performing this described technique which caused severe bruises, soft tissue damage, and metacarpal fractures.

The consequences are either avulsion amputations [144, 145] (Figs. 6.84 and 6.85), fractures or extensive soft tissue injuries (Fig. 6.86).

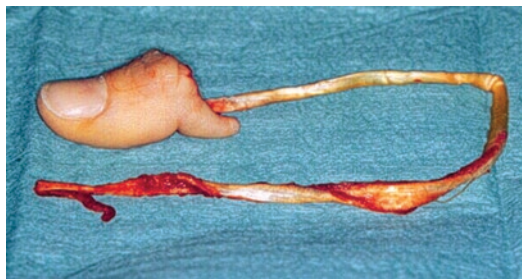


Fig. 6.84 Avulsion amputation through rope tangling in a climbers fall



Fig. 6.85 The finger could be replanted with interphalangeal fusion



Fig. 6.86 Climber fall injury caused by getting caught in a carabiner

The best treatment for these injuries is prevention, so climbers are advised to avoid grabbing the rope or carabiners during a fall [144]. In correct fall position, the hands should be away from the rope and used instead for stabilizing the body in the air during a fall [144] (Fig. 6.87) The therapy



Fig. 6.87 During a fall, the hands should be off the rope and used to stabilize the body in the air to avoid rope-tangling injury

of these rope-tangling injuries depends on their severity and follows the general orthopedic trauma guidelines. In the most minor cases, only rest and gentle range of motion exercises are required. In the most severe cases in which there is avulsion amputation of one or more digits, emergency surgical reconstruction in the form of digital replantation is required. Unfortunately, due to the wide zone of soft tissue injury in avulsion amputations, replanted digits often require further surgery, ranging from revision surgery to address severe stiffness to revision amputation surgery, if the digit necroses due to vascular compromise.

- **History:** Rope tangling during a fall or finger/hand getting crushed in a carabiner
- **Onset:** Acute
- **Clinical:** Dependent on trauma
- **Diagnostic:** Physical examination and x-ray
- **Therapy:** Dependent on extent of injury
- **Outcome:** Dependent on extent of injury



Fig. 6.88 Fracture of the fifth metacarpal in a climber caused by a fall while the hand was stuck in a crack in a long alpine climb. He finished the climb and a cast was applied the next day

6.3.17 Fractures

Most finger fractures in climbers are caused by direct trauma during rock fall or when hitting the rock during a fall. Another injury mechanism is an indirect trauma caused by a jammed hand or finger in a crack or pocket, causing non-axial forces or bending of the finger. Note that some minor fractures can present with the clinical symptoms of a pulley rupture; thus, in suspicion of a pulley rupture, an X-ray to exclude a fracture should be performed [32].

Finger fractures must be treated according to trauma/orthopedic surgical standards [19, 146, 147] (Fig. 6.88) specific to the fractured bone and the fracture pattern. These treatments vary from a short period of rest and gentle range of motion exercises for 4–6 weeks to surgery for reduction and hardware fixation of the fracture.

- **History:** Mostly direct trauma
- **Onset:** Acute
- **Clinical:** Deformity, pain while movement
- **Diagnostic:** Physical examination and X-ray, CT scan

- **Therapy:** Dependent on the kind of fracture either conservative or surgical
- **Outcome:** Dependent on extend of injury

6.3.18 Cartilage Compression Injury

Cartilage compression syndrome, or “over-crimping” injury, is an entity that has not been previously described in climbing literature but is now seen more frequently, probably because of the increasing difficulty level of climbing in recent years. We have found this condition to be associated with overuse of the crimping position. In this position, the PIP joint is almost completely flexed while the DIP joint is hyperextended. This causes stress points on the cartilage concentrated at certain small areas of the joint. Many pockets and small edges additionally require some tilting and twisting of the fingers, leading to shear forces onto the joints’ cartilage surface. These high-pressure peaks compress the cartilage and cause microinjuries. This causes a reaction of the joints’ inner synovial membrane which secretes fluid, leading to an articular effusion and pain. Long term, this injury could be a precursor to capsulitis and chronic synovitis of the IP joint. In its acute state, this condition causes pain when crimping, usually at the DIP joint. Open finger position moves are usually pain-free. Diagnosis is mostly based on the history and clinical examination, though MRI can also detect minor bone bruising and edema. The therapy consists mostly of avoidance of the triggering movement or complete rest from climbing until pain subsides completely. Additionally, icing, ointment dressings, and sometimes intra-articular injections with platelet-rich plasma (PRP) are performed. Oral supplementation with chondroitin and glucosamine sulfate also helps to nourish the cartilage and may be beneficial (Figs. 6.89 and 6.90).

- **History:** Hard crimping move
- **Onset:** Acute
- **Clinical:** Painful crimping test with sharp pain
- **Diagnostic:** History and clinical finding, maybe MRI
- **Therapy:** Conservative
- **Outcome:** Can result in chronic capsulitis



Fig. 6.89 Hard crimping puts high pressure onto the cartilage of the IP joints (Photo Michael Simon)

6.4 Conclusion

Modern climbing places tremendous unnatural strain on the joints and soft tissues of the fingers. Therefore, it is not surprising that the fingers are the most commonly injured body part in climbers, with more than 20 different climbing-related finger pathologies described to date. Several of these finger conditions, such as an acute closed rupture of the flexor pulley, repetitive stress epiphyseal fractures, and closed lumbrical muscle shear injuries, are interesting in that they are found almost exclusively in climbers.

Diagnosis of finger pain in climbers can usually be made with a detailed history of the injury, including the timeframe of injury, mechanism of injury, and aggravating and alleviating factors, such as specific gripping positions. Thereafter, a detailed examination of the hand, with injury-specific physical examination maneuvers, should help narrow down the diagnosis, which can then be confirmed with imaging techniques such as X-ray or dynamic ultrasound. MRI or CT scans



Fig. 6.90 Bone edema after hard crimping in the middle phalanx shown in the MRI

may be required in cases in which the diagnosis is still uncertain.

Treatment of climbing-related finger injuries is usually conservative, including a brief period of rest and anti-inflammatory measures, followed by focused physiotherapy and gradual return to sport-specific loading of the injured structure(s) with supportive taping as required. When conservative treatment fails, injectable and other minimally invasive treatments, such as steroid, PRP, stem cells, fat grafting, radial shock wave therapy, RSO, or leech therapy, may be beneficial. Surgical treatment is usually reserved for severe or specific conditions such as:

- Rope entanglement injuries
- Grossly displaced unstable fractures
- Unstable ligamentous injuries

- Complete combined closed flexor pulley ruptures
- Complete flexor or extensor tendon ruptures
- Non-united repetitive stress epiphyseal fractures
- Severe acute infections
- Chronic inflammatory conditions not resolved after exhausting nonsurgical treatment options
- Other non-climbing-specific hand surgical conditions such as arthritis, compressive neuropathies, ganglion cysts, tumors, and Dupuytren's contractures

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